

# Homological Algebra

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Bases on the lectures by Prof FEA Johnson

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# 1 Dates

13/01/2025

Notice this part is taken in 2025 so it might be different from this year.

**Definition 1.1** (Ring). A non-empty set  $R$  is a ring if it has two closed binary operations, addition and multiplication, satisfying the following conditions:

- (i)  $a + b = b + a$  for all  $a, b \in R$ .
- (ii)  $(a + b) + c = a + (b + c)$  for  $a, b, c \in R$ .
- (iii) There is an element  $0$  in  $R$  such that  $a + 0 = a$  for all  $a \in R$ .
- (iv) For every element  $a \in R$ , there exists an element  $-a$  in  $R$  such that  $a + (-a) = 0$ .
- (v)  $(ab)c = a(bc)$  for  $a, b, c \in R$ .
- (vi) For  $a, b, c \in R$ ,

$$a(b + c) = ab + ac$$

$$(a + b)c = ac + bc$$

If there is an element  $1 \in R$  such that  $1 \neq 0$  and  $1a = a1 = a$  for each element  $a \in R$ , we say that  $R$  is a ring with unity or identity. A ring  $R$  for which  $ab = ba$  for all  $a, b \in R$  is called a commutative ring. A commutative ring  $R$  with identity is called an integral domain if, for every  $a, b \in R$  such that  $ab = 0$ , either  $a = 0$  or  $b = 0$ . A division ring is a ring  $R$ , with an identity, in which every non-zero element in  $R$  is a unit; that is, for each  $a \in R$  with  $a \neq 0$ , there exists a unique element  $a^{-1}$  such that  $a^{-1}a = aa^{-1} = 1$ . A commutative division ring is called a field.

In first year, we consider vector spaces over fields:

$$\mathbb{F} \times V \rightarrow V$$

$$(\lambda, x) \mapsto \lambda \cdot x$$

where  $V$  is the abelian group. By a basis for  $V$ , we mean a collection of  $\langle e_1, \dots, e_n \rangle \in V$  which satisfies linear independence and Span. If  $T : V \rightarrow W$  is linear and  $\langle e_1, \dots, e_n \rangle$  is a basis for  $V$  and  $\langle \varphi_1, \dots, \varphi_m \rangle$  for  $W$  then  $T$  is represented by a matrix such that

$$T(e_i) = \sum_{j=1}^m \varphi_j a_{ji}$$

If  $R$  is not a field then everything collapses and how badly depends on  $R$ . One idea that remains is the idea of a free module.

**Definition 1.2.** A module over ring  $R$  is defined by  $(M, +, 0, \cdot)$  which satisfies

- (i)  $(M, +, 0)$  is an abelian group.
- (ii) The map  $\cdot : M \times R \rightarrow M$  satisfies

$$m \cdot (\lambda + \mu) = m \cdot \lambda + m \cdot \mu$$

$$(m \cdot \lambda) \cdot \mu = m \cdot (\lambda \cdot \mu)$$

$$m \cdot 1 = m$$

$$m \cdot 0 = 0$$

$$(m_1 + m_2) \cdot \lambda = m_1 \cdot \lambda + m_2 \cdot \lambda$$

**Definition 1.3.** If  $M$  is a module over  $R$  then  $R$  is free when it has a basis i.e. when  $\exists \{e_1, \dots, e_n\} \subset M$  such that  $\{e_1, \dots, e_n\}$  are linearly independent and  $M$  is generated by  $\{e_1, \dots, e_n\}$ . In particular,  $n$  is called the rank of  $M$ .

**Lemma 1.4.** If  $R$  is a ring in which each module is free, then  $R$  is a division ring. In particular, if given that  $R$  is commutative then  $R$  is a field.

**Proof.**

□

**Definition 1.5.** Let  $U, V$  be modules over  $R$ .  $R$ -linear map  $T : U \rightarrow V$  is defined by

$$T(x + y) = T(x) + T(y) \quad \forall x, y \in U$$

$$T(x\lambda) = T(x)\lambda \quad \forall x \in U, \lambda \in R$$

$$T(0) = 0$$

$\ker(T) = \{x \in U : T(x) = 0\}$  is a submodule of  $U$ .

Free resolutions. Given module  $M$  over ring  $R$ , we will assume  $M$  is finitely generated i.e.  $\exists \{m_1, \dots, m_n\} \subset M$  such that

$$\forall x \in M, x = m_1 \lambda_1 + \dots + m_n \lambda_n \quad \forall i \in \mathbb{R}$$

i.e. assume  $M$  has a finite spanning set but  $M$  is not free i.e. no spanning set is linear independent.

**Example.** Now take free module  $F = \mathbb{R}^n$  of rank  $n$  over  $R$  i.e.

$$F = \left\{ \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} : x_i \in R \right\}$$

$F$  has bases  $e_1, \dots, e_n$  the regular basis. We now construct linear map

$$\begin{aligned} \varphi : F &\rightarrow M \\ e_i &\mapsto m_i \end{aligned}$$

i.e.  $\varphi(\sum e_i \lambda_i) = \sum m_i \lambda_i$ .  $\varphi$  is surjective as  $m_1, \dots, m_n$  generates  $M$ . But if  $M$  is not free then  $\varphi$  cannot be injective and so  $\ker(\varphi)$  is non-trivial. So repeat the process for  $\ker(\varphi)$ . Let  $k_1, \dots, k_N$  be a generated set for  $\ker(\varphi)$ . Now take  $F_1$  be the free module on  $N$  generators

$$\begin{array}{ccccc} F_1 & \xrightarrow{\partial_1} & F_0 & \xrightarrow{\varphi} & M \longrightarrow 0 \\ & \searrow & \swarrow & & \\ & & \ker(\varphi) & & \end{array}$$

If  $\ker(\varphi)$  is free we get an exact sequence

$$0 \rightarrow F_1 \rightarrow F_0 \rightarrow M \rightarrow 0$$

If  $\ker(\varphi)$  is not free then  $\ker(\partial_1) \neq 0$ . Then we could have

$$\begin{array}{ccccccc} & & & \ker(\partial_1) & & & \\ & & \nearrow & \searrow & & & \\ F_3 & \xrightarrow{\quad} & F_2 & \xrightarrow{\partial_2} & F_1 & \xrightarrow{\partial_1} & F_0 \xrightarrow{\varphi} M \longrightarrow 0 \\ & \searrow & \nearrow & & \searrow & \swarrow & \\ & & \ker(\partial_2) & & & \ker(\varphi) & \end{array}$$

**Definition 1.6.** For a ring  $R$ , modules  $U, V, W$  over  $R$  a linear maps

$$U \xrightarrow{T} V \xrightarrow{S} W$$

Then sequence is exact at  $V$  when  $\text{Im}(T) = \ker(S)$ . More generally, if we have a sequence of linear maps

$$V_{n+2} \xrightarrow{f_{n+2}} V_{n+1} \xrightarrow{f_{n+1}} \dots \xrightarrow{f_0} 0$$

**Example.**

- (1) Every field is a commutative ring.
- (2)  $\mathbb{Z}$  is a ring which is not a field.
- (3) Consider  $\mathbb{Z}/2\mathbb{Z}$  given by  $\langle \bar{0}, \bar{1} \rangle$  then

$$\mathbb{Z} \xrightarrow{\times 2} \mathbb{Z} \xrightarrow{\varphi} \mathbb{Z}/2\mathbb{Z}$$

and  $\varphi$  is clearly surjective and  $\ker \varphi = 2\mathbb{Z}$ .

**Proposition 1.7.**

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\times 2} \mathbb{Z} \xrightarrow{\varphi} \mathbb{Z}/2\mathbb{Z}$$

is exact.

**Proof.**  $\varphi$  is surjective so exact at  $\mathbb{Z}/2\mathbb{Z}$ .  $\times 2$  is injective so exact here.

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\times 2} \mathbb{Z}$$

$\ker(\varphi) = 2\mathbb{Z}$  so exact here. Thus the overall is exact. □

**Remark.**

$$V \xrightarrow{f} W \longrightarrow 0$$

is exact if and only if  $f$  is surjective.

**Remark.**

$$0 \longrightarrow U \xrightarrow{g} V$$

is exact if and only if  $g$  is injective.

**Example.** Let  $R$  be a ring. By the polynomial ring  $R[x]$  has the expansion

$$a_0 + a_1x + \cdots + a_mx^m$$

where  $m \geq 0$  and  $a_i \in R$ . We make the convention of two polynomials are equal if all of the coefficients coincide. Let  $\alpha = a_0 + a_1x + \cdots + a_mx^m$  and  $\beta = b_0 + b_1x + \cdots + b_mx^m$  then

$$\alpha + \beta = (a_0 + b_0) + (a_1 + b_1)x + \cdots + (a_m + b_m)x^m$$

and

$$\alpha\beta = \sum c_n x^n \quad c_n = \sum_r a_r b_{n-r}$$

**Example.** Let  $R$  be a ring. Take  $R = \mathbb{Z}[x]$ , polynomial ring in a simple variable  $x$  over  $\mathbb{Z}$ . We can regard  $\mathbb{Z}$  as a module over  $\mathbb{Z}[x]$  by sending  $x$  to 0.

$$0 \longrightarrow \mathbb{Z}[y] \xrightarrow{\times y} \mathbb{Z}[y] \xrightarrow{\varphi} \mathbb{Z} \longrightarrow 0$$

and  $\varphi(\sum_{r \geq 0} a_r y^r) = a_0$ .

**Example.** Consider  $R[x, y] = (R[x])[y]$  the polynomial ring in two variables. Consider  $\mathbb{Z}$  as a module over  $\mathbb{Z}[x, y]$  by sending both  $x, y$  to 0.

$$0 \longrightarrow \mathbb{Z}[x, y] \longrightarrow \mathbb{Z}[x, y] \oplus \mathbb{Z}[x, y] \longrightarrow \mathbb{Z}[x, y] \xrightarrow{\varphi} \mathbb{Z} \longrightarrow 0$$

$\varphi(\sum a_{rs} x^r y^s) = a_{00}$ . This sequence is exact.

Koszul Resolution. If we regard  $\mathbb{Z}$  as a module over  $\mathbb{Z}[x_1, \dots, x_n]$  then we need  $n$  steps for a free resolute.

**Example.** Consider  $R = \mathbb{Z}(C_2)$  where  $C_2 = \{1, t\}$  with  $t^2 = 1$  so  $\mathbb{Z}(C_2) = \{a + bt \mid a, b \in \mathbb{Z}\}$  which is indeed a well-defined commutative ring. Regard  $\mathbb{Z}$  as a module over  $\mathbb{Z}(C_2)$

$$\mathbb{Z}[C_2] \xrightarrow{t-1} \mathbb{Z}[C_2] \xrightarrow{\varepsilon} \mathbb{Z} \longrightarrow 0$$

where  $\varepsilon(a + bt) = a + b$ . Note a module over  $\mathbb{Z}(C_2)$ ,  $\mathbb{Z}$  is infinite dimensional.

**Definition 1.8.**  $\text{Hom}_R(M, N) = \{f : M \rightarrow N \text{ is } R\text{-linear}\}$  and  $\text{Hom}_R(M, N)$  is an abelian group provided  $(f+g)(m) = f(m) + g(m)$  and  $0(m) = 0$ .

Suppose  $\alpha : M \rightarrow M'$  is  $R$ -linear then

$$\begin{array}{ccc} M & \xrightarrow{\alpha} & M' \\ & \searrow f \circ \alpha & \downarrow f \\ & & N \end{array}$$

$\alpha^* : \text{Hom}(M', N) \rightarrow \text{Hom}(M, N)$  with  $\alpha^*(f) = f \circ \alpha$ . Now suppose  $\beta : M' \rightarrow M''$  then

$$\begin{array}{ccccc} M & \xrightarrow{\alpha} & M' & \xrightarrow{\beta} & M'' \\ & \searrow g \circ \beta \circ \alpha & \downarrow g \circ \beta & \downarrow g & \\ & & N & & \end{array}$$

and

$$\begin{array}{ccccc} \text{Hom}(M'', N) & \xrightarrow{\beta^*} & \text{Hom}(M', N) & \xrightarrow{\alpha^*} & \text{Hom}(M, N) \\ & \searrow (\beta \circ \alpha)^* & & & \end{array}$$

Consider

$$\text{Hom}(M, N) \xrightarrow{\gamma_x} \text{Hom}(M, N')$$

where  $\gamma_x(f) = \gamma \circ f$  then  $\delta_x \gamma_x = (\delta \gamma)_x$ .

## 1.1 16/01/2026

The contravariant of  $\text{Hom}(\cdot, N)$ :

$$\begin{array}{ccccc} X & \xrightarrow{B} & Y & \xrightarrow{\alpha} & Z \\ & \searrow & & \searrow & \downarrow f \\ & & & f \circ \alpha & N \\ & & & \searrow & \\ & & & f \circ \alpha \circ \beta & \end{array}$$

Then  $\alpha^*(f) = f \circ \alpha$  and  $(\alpha \circ \beta)^* f = \beta^*(\alpha^*(f))$  where  $f \in \text{Hom}_\Lambda(Z, N)$ .

**Example.** If

$$\begin{array}{ccc} X & \xrightarrow{\text{id}} & X \\ & \searrow & \downarrow f \\ & & N \end{array}$$

Then  $\text{id}^*(f) = f$  so  $\text{id}^* = \text{id}$ .

Basic procedure: Let

$$\mathcal{F} = \left( \cdots F_{n+1} \xrightarrow{\partial_{n+1}} F_n \xrightarrow{\partial_n} \cdots \xrightarrow{\partial_2} F_1 \xrightarrow{\partial_1} F_0 \xrightarrow{\varepsilon} M \longrightarrow 0 \right)$$

Choose a module  $N$  and apply  $\text{Hom}_\Lambda(\cdot, N)$  to  $\mathcal{F}$  except the  $M$  term we get

$$0 \xrightarrow{\partial_1^*} \text{Hom}_\Lambda(F_1, N) \xrightarrow{\partial_2^*} \cdots \xrightarrow{\partial_{n-1}^*} \text{Hom}_\Lambda(F_{n-1}, N) \xrightarrow{\partial_n^*} \text{Hom}_\Lambda(F_n, N) \xrightarrow{\partial_{n+1}^*} \text{Hom}_\Lambda(F_{n+1}, N) \longrightarrow \cdots$$

Since  $\partial_n \circ \partial_{n+1} = 0$  in  $\mathcal{F}$ , applying  $\text{Hom}_\Lambda(\mathcal{F}, N)$  we get  $\partial_{n+1}^* \circ \partial_n^* = 0$ . However,  $\text{Hom}(\mathcal{F}, N)$  is not exact! But at least,  $\partial_n \circ \partial_{n+1} = 0$  implies  $\text{Im}(\partial_n^*) \subset \ker(\partial_{n+1}^*)$ . We have the following definition:

**Definition 1.9** (Cohomology). We define the  $n^{\text{th}}$  cohomology by

$$H_{\mathcal{F}}^n(M, N) = \ker \partial_{n+1}^* / \text{Im}(\partial_n^*).$$

**Example.** Let  $\Lambda = \mathbb{Z}$ ,  $M = \mathbb{Z}/p\mathbb{Z}$ . We have a free resolution of  $M$  over  $\mathbb{Z}$ :

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\times p} \mathbb{Z} \xrightarrow{\{\cdot\}} \mathbb{Z}/p\mathbb{Z} \longrightarrow 0$$

where  $\{\cdot\}$  is the canonical map. Take  $N = \mathbb{Z}$ , applying  $\text{Hom}_{\mathbb{Z}}(\cdot, \mathbb{Z})$  to  $\mathcal{F}$  we get

$$0 \longrightarrow \underbrace{\text{Hom}_{\mathbb{Z}}(\mathbb{Z}, \mathbb{Z})}_{=\mathbb{Z}} \xrightarrow{p^*} \text{Hom}_{\mathbb{Z}}(\mathbb{Z}, \mathbb{Z}) \longrightarrow 0$$

We see that  $(xp)^*(f)(x) = f(px) = pf(x)$ . Therefore,

$$H_{\mathcal{F}}^0(\mathbb{Z}, \mathbb{Z}) = 0, \quad H_{\mathcal{F}}^1(\mathbb{Z}, \mathbb{Z}) = \mathbb{Z}/p\mathbb{Z}, \quad H_{\mathcal{F}}^k(\mathbb{Z}, \mathbb{Z}) = 0 \text{ for all } k \geq 2$$

**Example.** Let  $C_2 = \{t \mid t^2 = 1\}$ ,  $Z(C_2) = \{a + bt \mid a, b \in \mathbb{Z}\}$ . Addition obvious and the multiplication is given by

$$(a + bt)(c + dt) = ac + bd + (ad + bc)t.$$

As  $M = \mathbb{Z}$ ,  $\mathbb{Z}$  acts trivially take  $\varepsilon = \text{id}$  then

$$\xrightarrow{1+t} Z(C_2) \xrightarrow{1-t} Z(C_2) \xrightarrow{1+t} Z(C_2) \xrightarrow{1-t} Z(C_2) \xrightarrow{\varepsilon} \mathbb{Z} \longrightarrow 0$$

Apply  $\text{Hom}_{Z(C_2)}(\cdot, \mathbb{Z})$  where  $\mathbb{Z}$  is the trivial module and the fact that  $\text{Hom}_\Lambda(\Lambda, N) \cong N$  we get

$$0 \longrightarrow \text{Hom}(Z(C_2), \mathbb{Z}) \xrightarrow{(1-t)^*} \text{Hom}(Z(C_2), \mathbb{Z}) \xrightarrow{(1+t)^*} \cdots$$

i.e.

$$0 \longrightarrow \mathbb{Z} \xrightarrow{0} \mathbb{Z} \xrightarrow{\times 2} \mathbb{Z} \xrightarrow{0} \cdots$$

since  $(1-t)^* = 0$  and  $(1+t)^* = 2$ . In this case we have

$$H_{Z(C_2)}^0(\mathbb{Z}, \mathbb{Z}) = 0, \quad H_{Z(C_2)}^{2k-1}(\mathbb{Z}, \mathbb{Z}) = 0, \quad H_{Z(C_2)}^{2k}(\mathbb{Z}, \mathbb{Z}) = \mathbb{Z}/2\mathbb{Z} \text{ for } k \in \mathbb{N}$$

**Proposition 1.10.** Let  $\Lambda$  be a ring,  $N$  be a  $\Lambda$ -module then

$$\text{Hom}_\Lambda(\Lambda, N) \cong N.$$

**Proof.** Let  $f : \Lambda \rightarrow N$  be completely determined by  $f(1)$  i.e.  $f(\lambda) = f(1)\lambda$ . Then

$$\begin{aligned} \nu : \text{Hom}_\Lambda(\Lambda, N) &\rightarrow N \\ f &\mapsto f(1) \end{aligned}$$

is an isomorphism. □

Procedure:

1. Given  $M$ , we need to construct a free resolution of  $M/\Lambda$ :

$$\mathcal{F} = \left( \cdots F_{n+1} \xrightarrow{\partial_{n+1}} F_n \xrightarrow{\partial_n} \cdots \xrightarrow{\partial_2} F_1 \xrightarrow{\partial_1} F_0 \xrightarrow{\varepsilon} M \longrightarrow 0 \right)$$

2. Given a coefficient module  $N$ , compute

$$H_{\mathcal{F}}^n(M, N) = \frac{\ker \partial_{n+1}^* : \text{Hom}(F_n, N) \rightarrow \text{Hom}(F_{n+1}, N)}{\text{Im} \partial_n^* : \text{Hom}(F_{n-1}, N) \rightarrow \text{Hom}(F_n, N)}$$

We need to show that

3.  $H_{\mathcal{F}}^n(M, N)$  is independent of free resolution chosen.
4. Same holds for "projective resolutions".  $P$  is projective if and only if  $H^1(P, N) = 0$  for all  $N$ .

## 1.2 19/01/2026

**Definition 1.11.** A module  $F$  is called projective when there exists  $\tilde{\varphi}$  such that the following diagram commutes:

$$\begin{array}{ccc} & F & \\ \swarrow \exists \tilde{\varphi} & \downarrow \varphi & \\ X & \xrightarrow{\eta} & M \end{array}$$

where  $\eta$  is surjective. In this case,  $\eta \circ \tilde{\varphi} = \varphi$ .

**Proposition 1.12.** Every free module is projective.

**Proof.** Take basis  $\{e_i\}_{i \in I}$  for  $F$ . As  $\eta$  is surjective for all  $i$ , choose  $x_i \in X_i$  such that  $\eta(x_i) = \varphi(e_i)$ . Let  $\tilde{\varphi} : F \rightarrow X$  via  $\tilde{\varphi}(e_i) =: x_i$  then this is exactly what we want. □

**Remark.** We will eventually produce an example that the converse is not true.

Lifting. Let  $P_* \rightarrow M$ ,  $Q_* \rightarrow M'$  be surjective projective resolution

$$\begin{array}{ccccccccccccccc} P_* = ( & \longrightarrow & P_{n+1} & \xrightarrow{\partial_{n+1}} & P_n & \xrightarrow{\partial_n} & \cdots & \xrightarrow{\partial_2} & P_1 & \xrightarrow{\partial_1} & P_0 & \xrightarrow{\varepsilon} & M & \longrightarrow & 0) \\ & & \downarrow f_{n+1} & & \downarrow f_n & & & & \downarrow f_1 & & \downarrow f_0 & & \downarrow f & & \\ Q_* = ( & \longrightarrow & Q_{n+1} & \xrightarrow{\delta_{n+1}} & Q_n & \xrightarrow{\delta_n} & \cdots & \xrightarrow{\delta_2} & Q_1 & \xrightarrow{\delta_1} & Q_0 & \xrightarrow{\varepsilon'} & M' & \longrightarrow & 0) \end{array}$$

Both sequences are exact,  $P_i, Q_j$  are projective but we can take it to be free.

**Definition 1.13.** By a lifting of  $f$  to  $P_* \rightarrow Q_*$ . A collection of homomorphisms  $(f_r)_{r \in \mathbb{N}}$ ,  $f_r : P_r \rightarrow Q_r$  such that the diagram above commutes i.e.

$$f \circ \varepsilon = \varepsilon' \circ f_0, \quad f_{n-1} \circ \partial_n = \delta_n \circ f_n$$

**Proposition 1.14.** Let  $P_* \rightarrow M$ ,  $Q_* \rightarrow M'$  be surjective projective resolutions. Let  $f : M \rightarrow M'$  be a  $\Lambda$ -homomorphism. Then there exists a lifting  $(f_r)_{r \in \mathbb{N}}$  of  $f$  to  $(f_r) : P_* \rightarrow Q_*$ .

**Proof.** We first construct  $f_0 : P_0 \rightarrow Q_0$ . Notice

$$\begin{array}{ccccc} P_0 & \xrightarrow{\varepsilon} & M & \longrightarrow & 0 \\ & & \downarrow f & & \\ Q_0 & \xrightarrow{\varepsilon_0} & M' & \longrightarrow & 0 \end{array}$$

□

By applying surjectivity, there exists some  $f_0$  such that the following diagram

$$\begin{array}{ccc} & P_0 & \\ \swarrow f_0 & \downarrow f \circ \varepsilon & \\ Q_0 & \xrightarrow{\varepsilon'} & M' \end{array}$$

commutes. This looks like

$$\begin{array}{ccccc} P_0 & \xrightarrow{\varepsilon} & M & \longrightarrow & 0 \\ \downarrow f_0 & & \downarrow f & & \\ Q_0 & \xrightarrow{\varepsilon_0} & M' & \longrightarrow & 0 \end{array}$$

By induction, suppose we have

$$\begin{array}{ccccc} P_n & \xrightarrow{\partial_n} & P_{n-1} & \xrightarrow{\partial_{n-1}} & P_{n-2} \\ & & \downarrow f_{n-1} & & \downarrow f_{n-2} \\ Q_n & \xrightarrow{\delta_n} & Q_{n-1} & \xrightarrow{\delta_{n-1}} & Q_{n-2} \end{array}$$

i.e.  $\delta_{n-1}f_{n-1}\partial_n = f_{n-2}\partial_{n-1}\partial_n$ . By exactness,  $\delta_{n-1}f_{n-1}\partial_n = 0$ . This implies  $\text{Im}(f_{n-1}\partial_n) \subset \ker \delta_{n-1}$ . By exactness,  $\ker \delta_{n-1} = \text{Im}(\delta_n)$ . Thus, by projectivity, there exists  $f_n$  such that

$$\begin{array}{ccc} & P_n & \\ f_n \swarrow & & \downarrow f_{n-1} \circ \partial_n \\ Q_n & \xrightarrow{\delta_n} & \text{Im}(\delta_n) \end{array}$$

commutes. And  $\delta_n \circ f_n = f_{n-1} \circ \partial_n$  i.e.

$$\begin{array}{ccc} P_n & \xrightarrow{\partial_n} & P_{n-1} \\ \downarrow f_n & & \downarrow f_{n-1} \\ Q_n & \xrightarrow{\delta_n} & Q_{n-1} \end{array}$$

commutes.

**Proposition 1.15.** Let  $f : M \rightarrow M'$  be a  $\Lambda$ -homomorphism and  $P_* \rightarrow M$ ,  $Q_* \rightarrow M'$  be surjective projective resolutions. Suppose we have a lifting  $(f_r)_{r \in \mathbb{N}}$  from  $P_* \rightarrow Q_*$ . Then  $(f_r)$  induces homomorphism

$$(f_r)^* : H^k(M', N) \rightarrow H^k(M, N) \quad \forall k \geq 0.$$

**Proof.** We have commutative diagram

$$\begin{array}{ccccccccccccccc} P_* = ( & \longrightarrow & P_{n+1} & \xrightarrow{\partial_{n+1}} & P_n & \xrightarrow{\partial_n} & \cdots & \xrightarrow{\partial_2} & P_1 & \xrightarrow{\partial_1} & P_0 & \xrightarrow{\varepsilon} & M & \longrightarrow & 0) \\ & & \downarrow f_{n+1} & & \downarrow f_n & & & & \downarrow f_1 & & \downarrow f_0 & & \downarrow f & & \\ Q_* = ( & \longrightarrow & Q_{n+1} & \xrightarrow{\delta_{n+1}} & Q_n & \xrightarrow{\delta_n} & \cdots & \xrightarrow{\delta_2} & Q_1 & \xrightarrow{\delta_1} & Q_0 & \xrightarrow{\varepsilon'} & M' & \longrightarrow & 0) \end{array}$$

Applying  $\text{Hom}_\Lambda(\cdot, N)$  we get

$$\begin{array}{ccccc} \text{Hom}_\Lambda(Q_{n+1}, N) & \xrightarrow{\delta_{n+1}^*} & \text{Hom}_\Lambda(Q_n, N) & \xrightarrow{\delta_n^*} & \text{Hom}_\Lambda(Q_{n-1}, N) \\ \downarrow f_{n+1}^* & & \downarrow f_n^* & & \downarrow f_{n-1}^* \\ \text{Hom}_\Lambda(P_{n+1}, N) & \xrightarrow{\partial_{n+1}^*} & \text{Hom}_\Lambda(P_n, N) & \xrightarrow{\partial_n^*} & \text{Hom}_\Lambda(P_{n-1}, N) \end{array}$$

We claim that  $f_n^* : \ker(\delta_{n+1}^*) \rightarrow \ker(\partial_{n+1}^*)$ . Since  $f_{n+1}^* \circ \delta_{n+1}^* = \partial_{n+1}^* \circ f_n^*$ . Let  $z \in \ker(\delta_{n+1}^*)$  then  $\delta_{n+1}^*(z) = 0$ . This implies  $\partial_{n+1}^*(f_n^*(z)) = 0$  so  $f_n^*(z) \in \ker(\partial_{n+1}^*)$ . We claim also that  $f_n^* : \text{Im}(\delta_n^*) \rightarrow \text{Im}(\partial_n^*)$ . Since  $f_n^* \circ \delta_n^* = \partial_n^* \circ f_{n-1}^*$ , let  $u \in \text{Im}(\delta_n^*)$  then clearly  $f_n^*(u) \in \text{Im}(\partial_n^*)$ . Therefore

$$\begin{aligned} f_n^* : \ker(\delta_{n+1}^*) &\rightarrow \ker(\partial_{n+1}^*) \\ f_n^* : \text{Im}(\delta_n^*) &\rightarrow \text{Im}(\partial_n^*) \end{aligned}$$

So it induces

$$f_n^* : \ker(\delta_{n+1}^*)/\text{Im}(\delta_n^*) \rightarrow \ker(\partial_{n+1}^*)/\text{Im}(\partial_n^*)$$

i.e.

$$f_n^* : H_Q^n(M', N) \rightarrow H_P^n(M, N).$$

□

To summarize,  $P_* \rightarrow M$ ,  $Q_* \rightarrow M'$  and  $f : M \rightarrow M'$  a  $\Lambda$ -homomorphism. We construct a lifting  $\{f_r\}_{r \in \mathbb{N}} : P_* \rightarrow Q_*$ . When we are applying  $\text{Hom}_\Lambda(\cdot, N)$ , we get an induced homomorphism

$$(f_r)^* : LH_Q^n(M', N) \rightarrow H_P^n(M, N)$$

We need to show that if  $(f'_r)_{r \in \mathbb{N}}$  is another lifting then it induces the same map, i.e.

$$(f'_r)^* = (f_r)^* : H_Q^n(M', N) \rightarrow H_P^n(M, N)$$

i.e. independent of the choice of lifting.

Special case: Consider a lifting  $(f_r)_{r \in \mathbb{N}}$  of the zero map  $0 : M \rightarrow M'$ , we claim that

$$(f_r)^* : H_Q^n(M', N) \rightarrow H_P^n(M, N).$$

Clear trick: (Null homotopy)

$$\begin{array}{ccccccccccccccc} P_* = ( & \longrightarrow & P_{n+1} & \xrightarrow{\partial_{n+1}} & P_n & \xrightarrow{\partial_n} & \cdots & \xrightarrow{\partial_2} & P_1 & \xrightarrow{\partial_1} & P_0 & \xrightarrow{\varepsilon} & M & \longrightarrow & 0) \\ & & \downarrow f_{n+1} & \swarrow \mathcal{H}_n & \downarrow f_n & & & & \downarrow f_1 & \swarrow \mathcal{H}_0 & \downarrow f_0 & & \downarrow 0 & & \\ Q_* = ( & \longrightarrow & Q_{n+1} & \xrightarrow{\delta_{n+1}} & Q_n & \xrightarrow{\delta_n} & \cdots & \xrightarrow{\delta_2} & Q_1 & \xrightarrow{\delta_1} & Q_0 & \xrightarrow{\varepsilon'} & M' & \longrightarrow & 0) \end{array}$$

Given the commutative diagram. We claim that there exists  $(\mathcal{H}_n) : P_n \rightarrow Q_{n+1}$ ,  $n \in \mathbb{N}$ . For  $n \geq 1$ ,  $f_n = \delta_{n+1}\mathcal{H}_n + \mathcal{H}_{n-1}\partial_n$

and  $f_0 = \delta_1 \mathcal{H}_0$ . First construct  $\mathcal{H}_0 : P_0 \rightarrow Q_1$  such that  $\delta_1 \mathcal{H}_0 = f_0$ . Notice  $\varepsilon' \circ f_0 = 0 \circ \varepsilon = 0$  so  $\text{Im}(f_0) \subset \ker(\varepsilon') = \text{Im}(\delta_1)$ . Thus, by surjectivity

$$\begin{array}{ccc} & P_0 & \\ \swarrow \mathcal{H}_0 & \downarrow f_0 & \\ Q_1 & \xrightarrow{\delta_1} & \text{Im}(\delta_1) \end{array}$$

such  $\mathcal{H}_0$  exists. Suppose inductively we have

$$\begin{array}{ccccccc} P_n & \xrightarrow{\partial_n} & P_{n-1} & \xrightarrow{\partial_{n-1}} & P_{n-2} & & \\ \downarrow f_n & \swarrow \mathcal{H}_{n-1} & \downarrow f_{n-1} & \swarrow \mathcal{H}_{n-2} & \downarrow f_{n-2} & & \\ Q_{n+1} & \longrightarrow & Q_n & \xrightarrow{\delta_n} & Q_{n-1} & \xrightarrow{\delta_{n-1}} & Q_{n-2} \end{array}$$

and  $f_{n-1} = \mathcal{H}_{n-2} \partial_{n-1} + \delta_n \mathcal{H}_{n-1}$ . Consider

$$\begin{aligned} \delta_n(f_n - \mathcal{H}_{n-1} \partial_n) &= \delta_n f_n - \delta_n \mathcal{H}_{n-1} \partial_n = f_{n-1} \partial_n - \delta_n \mathcal{H}_{n-1} \partial_n \\ &= (f_{n-1} - \delta_n \mathcal{H}_{n-1}) \partial_n = (\mathcal{H}_{n-2} \partial_{n-1}) \partial_n = 0 \end{aligned}$$

Thus,  $\text{Im}(f_n - \mathcal{H}_{n-1} \partial_n) \subset \ker(\partial_n) = \text{Im}(\delta_{n+1})$ . So we have

$$\begin{array}{ccc} & P_n & \\ \swarrow \mathcal{H}_n & \downarrow f_n - \mathcal{H}_{n-1} \partial_n & \\ Q_{n+1} & \xrightarrow{\delta_{n+1}} & \text{Im}(\delta_{n+1}) \end{array}$$

Then  $\delta_{n+1} \mathcal{H}_n = f_n - \mathcal{H}_{n-1} \partial_n$  i.e.  $f_n = \delta_{n+1} \mathcal{H}_n + \mathcal{H}_{n-1} \partial_n$ .

**Proposition 1.16.** If a lifting  $(f_r)_{r \in \mathbb{N}}$  has a null-homotopy  $(\mathcal{H}_n) : P_n \rightarrow Q_{n+1}$  then  $(f_n)^* : H_Q^n(M', N) \rightarrow H_P^n(M, N)$  is the zero map.

**Proof.** Since  $f_n = \delta_{n+1} \mathcal{H}_n + \mathcal{H}_{n-1} \partial_n$ . Applying  $\text{Hom}(\cdot, N)$  we have  $f_n^* = \mathcal{H}_n^* \delta_{n+1}^* + \partial_n^* \mathcal{H}_{n-1}^*$ . We claim  $f_n^* : \ker \delta_{n+1}^* \rightarrow \text{Im}(\partial_n^*)$ . Let  $z \in \ker \delta_{n+1}^*$ ,  $\delta_{n+1}^*(z) = 0$ . Then  $f_n^*(z) = \partial_n^*(\mathcal{H}_{n-1}^*(z)) \in \text{Im}(\partial_n^*)$ . But  $\text{Im}(\partial_n^*)$  defines zero in  $H_P^n(M, N)$ .  $\square$

**Corollary 1.17.** Let  $(f_r)_{r \in \mathbb{N}}, (f'_r)_{r \in \mathbb{N}}$  be two lifting from  $(f_r) : P_* \rightarrow Q_*$ ,  $(f'_r) : P_* \rightarrow Q_*$  then  $f_r^* = f'_r^*$ .

**Proof.**  $(f_r - f'_r)_{r \geq 0}$  is a lifting of zero map so  $(f_r^* - f'_r^*) \equiv 0$  i.e.  $f_r^* = f'_r^*$ .  $\square$

**Corollary 1.18.** Let  $P_* \rightarrow M, Q_* \rightarrow M$  be projective resolutions of  $M$ . Then

$$H_P^n(M, N) \cong H_Q^n(M, N)$$

i.e. the choice of projective resolutions is irrelevant. You get the same things up to isomorphism.

**Proof.**

$$\begin{array}{ccccccccccc} \longrightarrow & P_{n+1} & \xrightarrow{\partial_{n+1}} & P_n & \xrightarrow{\partial_n} & \cdots & \xrightarrow{\partial_2} & P_1 & \xrightarrow{\partial_1} & P_0 & \xrightarrow{\varepsilon} & M & \longrightarrow & 0 \\ & \downarrow f_{n+1} & & \downarrow f_n & & & & \downarrow f_1 & & \downarrow f_0 & & \downarrow \text{id} & & \\ \longrightarrow & Q_{n+1} & \xrightarrow{\delta_{n+1}} & Q_n & \xrightarrow{\delta_n} & \cdots & \xrightarrow{\delta_2} & Q_1 & \xrightarrow{\delta_1} & Q_0 & \xrightarrow{\varepsilon'} & M & \longrightarrow & 0 \\ & \uparrow g_{n+1} & & \uparrow g_n & & & & \uparrow g_1 & & \uparrow g_0 & & \uparrow \text{id} & & \\ \longrightarrow & P_{n+1} & \xrightarrow{\partial_{n+1}} & P_n & \xrightarrow{\partial_n} & \cdots & \xrightarrow{\partial_2} & P_1 & \xrightarrow{\partial_1} & P_0 & \xrightarrow{\varepsilon} & M & \longrightarrow & 0 \end{array}$$

$(f_r), (g_r)$  are two lifting from  $\text{id}_M : M \rightarrow M$ . Both  $(g_r \circ f_r)$  and  $(f_r \circ g_r)$  are lifting of  $\text{id} : M \rightarrow M$  from  $P$  to  $P$  and  $\text{id} : M \rightarrow M$  from  $Q$  to  $Q$  respectively. There is an obvious lifting  $(f'_r)$  of  $\text{id}$  from  $P$  to  $P$ . By uniqueness,

$$\text{id} = (g_r \circ f_r)^* : H_P^n(M, N) \rightarrow H_P^n(M, N).$$

Then  $(g_r \circ f_r)^* = f_r^* \circ g_r^* = \text{id} : H_P^n(M, N) \rightarrow H_P^n(M, N)$  so  $(f_r)^*$  and  $(g_r)^*$  are isomorphic.  $\square$

**Definition 1.19.** A chain complex is a sequence

$$C_{n+1} \xrightarrow{\partial_{n+1}} C_n \xrightarrow{\partial_n} \cdots$$

where  $\partial_n \partial_{n+1} = 0$ .

Applying  $\text{Hom}(\cdot, N)$  to a chain complex we get

$$\text{Hom}(C_{n-1}, N) \xrightarrow{\partial_n^*} \text{Hom}(C_n, N) \xrightarrow{\partial_{n+1}^*} \text{Hom}(C_{n+1}, N)$$

where  $\partial_{n+1}^* \partial_n^* = 0$ . Let  $C^n = \text{Hom}(C_n, N)$  then we have the following definition:

**Definition 1.20.**  $(C^n, \delta^n)_{n \in \mathbb{N}}$  is a cochain complex when each  $C^n$  is a  $\Lambda$ -module.

$$C^{n-1} \xrightarrow{\delta^{n-1}} C^n \xrightarrow{\delta^n} C^{n+1}$$

each  $\delta^r : C^r \rightarrow C^{r+1}$  is a  $\Lambda$ -homomorphism and  $\delta^n \circ \delta^{n-1} = 0$ . The cohomology of a cochain complex is

$$H^n(C^*) = \ker \delta^n / \text{Im}(\delta^{n-1})$$

If we define  $D_{-n} = C^n$  then

$$D_{-(n+1)} \longrightarrow D_{-n} \longrightarrow D_{-n+1}$$

Hence, there's no central difference of chain complex and cochain complex.

## 1.3 23/01/2026

Projective modules/splitting criteria.

**Lemma 1.21** (Five Lemma).

$$\begin{array}{ccccccccc} 0 & \longrightarrow & A & \longrightarrow & B & \longrightarrow & C & \longrightarrow & 0 \\ & & \downarrow \alpha & & \downarrow \beta & & \downarrow \gamma & & \\ 0 & \longrightarrow & A' & \longrightarrow & B' & \longrightarrow & C' & \longrightarrow & 0 \end{array}$$

commutative diagram of  $\Lambda$ -modules in which both modules are exact. Then  $\alpha$  and  $\gamma$  are isomorphisms implies  $\beta$  is isomorphism.

Let  $A, C$  be  $\Lambda$ -modules. Then we have a trivial exact sequence

$$0 \longrightarrow A \xrightarrow{i} A \oplus C \xrightarrow{\pi} C \longrightarrow 0$$

**Definition 1.22.** Let

$$0 \longrightarrow A \xrightarrow{j} B \xrightarrow{p} C \longrightarrow 0$$

be exact. We say this splits when there exists a commutative diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & A & \longrightarrow & B & \longrightarrow & C & \longrightarrow & 0 \\ & & \downarrow \text{id}_A & & \downarrow \varphi & & \downarrow \text{id}_C & & \\ 0 & \longrightarrow & A & \longrightarrow & A \oplus C & \xrightarrow{\pi} & C & \longrightarrow & 0 \end{array}$$

where

$$j(a) = \begin{pmatrix} a \\ 0 \end{pmatrix}, \quad \pi \begin{pmatrix} a \\ c \end{pmatrix} = c$$

Then by Five Lemma,  $\varphi$  is an isomorphism.

We say that

$$\varepsilon = (0 \longrightarrow A \xrightarrow{j} B \xrightarrow{p} C \longrightarrow 0)$$

splits in the right when there exists a  $\Lambda$ -homomorphism  $s : C \rightarrow B$  such that  $p \circ s = \text{id}_C$ . We show that "splits on right" implies splits on the left.  $\varepsilon$  splits on left, if there exists  $\nu : B \rightarrow A$  ( $\Lambda$ -homomorphism) such that  $\nu \circ j = \text{id}_A$ .

**Exercise.**  $\varepsilon$  "splits" if and only if  $\varepsilon$  "splits on right" and if and only  $\varepsilon$  "splits on the left".

We shall prove "splits on right" implies "splits".

**Proposition 1.23.**

$$\varepsilon = (0 \longrightarrow A \xrightarrow{j} B \xrightarrow{p} C \longrightarrow 0)$$

be an exact sequence of  $\Lambda$ -modules. Then  $\varepsilon$  splits right implies  $\varepsilon$  splits.

**Proof.** Suppose  $s : C \rightarrow B$  satisfies  $p \circ s = \text{id}_C$ . Consider the diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & A & \xrightarrow{i} & A \oplus C & \xrightarrow{\pi} & C & \longrightarrow & 0 \\ & & \downarrow \text{id}_A & & \downarrow \varphi & & \downarrow \text{id}_C & & \\ 0 & \longrightarrow & A & \xrightarrow{j} & B & \xrightarrow{p} & C & \longrightarrow & 0 \end{array}$$

Define  $\varphi : A \oplus C \rightarrow B$  by

$$\varphi \begin{pmatrix} a \\ c \end{pmatrix} = j(a) + s(c)$$

then

$$\begin{array}{ccc} A \oplus C & \xrightarrow{\pi} & C \\ \downarrow \varphi & & \downarrow \text{id}_C \\ B & \xrightarrow{p} & C \end{array}$$

commutes. Then

$$p(\varphi \begin{pmatrix} a \\ c \end{pmatrix}) = p(j(a) + s(c)) = pj(a) + ps(c)$$

But  $p \circ j = 0$  due to exactness. Thus,

$$p(\varphi \begin{pmatrix} a \\ c \end{pmatrix}) = p \circ s(c) = c = \pi \begin{pmatrix} a \\ c \end{pmatrix}$$

$$\begin{array}{ccc} A & \xrightarrow{i} & A \oplus C \\ \downarrow \text{id} & & \downarrow \varphi \\ A & \xrightarrow{j} & B \end{array}$$

also commutes. So  $\varphi$  is an isomorphism by Five Lemma. SO  $\varphi^{-1}$  exists i.e.  $\varepsilon$  splits. □

Projective modules definitions:

1. Suppose given a diagram of  $\Lambda$ -modules

$$\begin{array}{ccc} & & P \\ & \swarrow \tilde{\varphi} & \downarrow \varphi \\ X & \xrightarrow{p} & M \end{array}$$

$P$  is a projective when there exists  $\tilde{\varphi} : P \rightarrow X$  such that  $p \circ \tilde{\varphi} = \varphi$  for  $p$  surjective.

2. Suppose given exact sequence of  $\Lambda$ -modules

$$0 \longrightarrow K \xrightarrow{i} X \xrightarrow{p} P \longrightarrow 0$$

$P$  is projective if and only if every such exact sequence splits.

3.  $P$  is projective if and only if there exists  $\Lambda$ -module  $K$  such that  $K \oplus P$  is free.

4.  $P$  is projective if and only if  $H^1(P, N) = 0$  for any  $N$ .

**Proposition 1.24.** All the above definitions are equivalent. (In particular, the first three of them).

**Proof.** (1  $\implies$  2) : Given an exact sequence

$$\varepsilon = (0 \longrightarrow K \xrightarrow{i} X \xrightarrow{p} P \longrightarrow 0)$$

$$\begin{array}{ccc} & & P \\ & \swarrow s=\tilde{\text{id}} & \downarrow \text{id} \\ X & \xrightarrow{p} & P \end{array}$$

By definition 1. There exists  $s : P \rightarrow X$  such that  $p \circ s = \text{id}_P$  i.e.  $\varepsilon$  splits on right so  $\varepsilon$  splits.

(2  $\implies$  3) : Let  $P$  be a module and let  $(e_i)_{i \in I}$  be a set of generators for  $B$  i.e. for all  $x \in P$ , there exists  $(x_i)_{i \in I}$ ,  $x_i \in \Lambda$  such that  $x = \sum e_i x_i$ . Let  $(E_i)_{i \in I}$  be a basis for the free module  $F$ . Consider  $p : F \rightarrow P$  given by  $p(E_i) = e_i$  which is clearly surjective so we have an exact sequence

$$0 \longrightarrow K \xrightarrow{i} X \xrightarrow{p} P \longrightarrow 0$$

where  $K = \ker(p)$ . By definition 2, this splits. So  $F \subseteq K \oplus P$ . i.e. there exists  $K$  such that  $K \oplus P$  is free.

(3  $\implies$  1) : Given a diagram

$$\begin{array}{ccc} & & P \\ & \swarrow \tilde{\varphi} & \downarrow \varphi \\ X & \xrightarrow{p} & M \end{array}$$

we need to produce  $\tilde{\varphi}$ . By definition 3, there exists  $K$  such that  $P \oplus K$  is free. Take a basis  $(E_i)_{i \in I}$  for  $P \oplus K$ . Modify  $\varphi$  as following. Define  $\psi : P \oplus K \rightarrow M$  by

$$\psi \begin{pmatrix} x \\ y \end{pmatrix} = \varphi(x) \quad x \in P, y \in K$$

Because  $p : X \rightarrow M$  is surjective for each  $i \in I$ . Choose  $\xi_i \in X$  such that  $p(\xi_i) = \psi(E_i)$ . Define  $\tilde{\psi} : P \oplus K \rightarrow X$  by  $\tilde{\psi}(E_i) = \xi_i$ . Define  $\tilde{\varphi} : P \rightarrow X$  by

$$\tilde{\varphi}(x) = \tilde{\psi} \begin{pmatrix} x \\ 0 \end{pmatrix}$$

$$p\tilde{\varphi} = p\tilde{\psi} \begin{pmatrix} x \\ 0 \end{pmatrix} = \psi \begin{pmatrix} x \\ 0 \end{pmatrix} = x.$$

□

**Example.** Let  $R, S$  be map.

$R \times S = \{(x, y) \mid x \in R, y \in S\}$  is also a ring

So  $R \times S$  has two obvious (non free) projective module.  $P_1 = R \times \{0\}$ ,  $P_2 = \{0\} \times S$ .  $P_1 \oplus P_2 = R \times S$  is free of rank 1. By the Theorem in Representation Theory. If  $\mathbb{F}$  be algebraically closed and  $\text{char}(\mathbb{F}) = 0$  then

$$\mathbb{F}[G] = \prod_{i=1}^m M_{d_i}(\mathbb{F})$$

$m$  is the number of conjugacy class of  $G$ . In particular

$$\sum d_i^2 = |G|$$

Another consequence is Every module over  $\mathbb{F}(G)$  is projective.

## 1.4 26/01/2026

A brief summary: Let  $M$  be a  $\Lambda$ -module, we take  $P_* \rightarrow M$  be a projective resolution. We take  $N$  be the coefficient module, we take its cochain complex  $\text{Hom}(P_*, N)$ , its cohomology is given by  $H_\Lambda^n(M, N) = \ker \partial_{n+1}^* / \text{Im}(\partial_n^*)$  which is independent of the particular particles  $P_*$ .

**Proposition 1.25.**  $H^0(M, N) \cong \text{Hom}_\Lambda(M, N)$ .

**Proof.** First few terms of  $P_*$ :

$$\begin{array}{ccccccc} P_1 & \xrightarrow{\partial_1} & P_0 & \xrightarrow{\varepsilon} & M & \longrightarrow & 0 \\ & & & \searrow \varepsilon^*(\alpha) & \downarrow \alpha & & \\ & & & & N & & \end{array}$$

First notice that  $\varepsilon^*(\alpha) \in \text{Hom}(P_0, N)$ , but actually  $\varepsilon^*(\alpha) \in \ker(\partial_1^*)$ . Indeed,  $\partial_1^*(\varepsilon^*(\alpha)) = \alpha \circ \varepsilon \circ \partial_1 = 0$  since  $\varepsilon \circ \partial_1 = 0$ . Hence

$$\varepsilon^* : \text{Hom}(M, N) \rightarrow \ker(\partial_1^*)$$

By definition of  $H^0$ :

$$H^0(M, N) = \ker(\partial_1^* : \text{Hom}(P_0, N) \rightarrow \text{Hom}(P_1, N))$$

We claim  $\varepsilon^* : \text{Hom}_\Lambda(M, N) \rightarrow H^0(M, N)$  is bijective.

(Injectivity): Suppose  $\varepsilon^*(\alpha) = 0$  then  $\alpha \circ \varepsilon = 0$ . We claim that  $\alpha = 0$ . Since  $\varepsilon$  is surjective, if we take  $m \in M$  there exists some  $x \in P_0$  such that  $\varepsilon(x) = m$ . This implies  $\alpha(m) = \alpha \circ \varepsilon(x) = 0$  so  $\alpha(m) = 0$  for all  $m$  i.e.  $\alpha = 0$ .

(Surjectivity):  $\varepsilon^* : \text{Hom}_\Lambda(M, N) \rightarrow \ker \partial_1^*$  is surjective. Let  $\alpha \in \text{Hom}(P_0, N)$ , this implies  $\partial_1^*(\alpha) = \alpha \circ \partial_1 = 0$  i.e.  $\alpha(\text{Im}(\partial_1)) = 0$ .  $\text{Im}(\partial_1) = \ker(\varepsilon)$  by exactness. Thus

$$\begin{array}{ccccccc} 0 & \longrightarrow & \ker(\varepsilon) & \longrightarrow & P_0 & \xrightarrow{\varepsilon} & M \longrightarrow M \\ & & & & \downarrow \alpha & & \\ & & & & N & & \end{array}$$

$\alpha$  induces a mapping  $\tilde{\alpha} : P_0 / \ker(\varepsilon) \rightarrow M$  via  $\tilde{\alpha}(x + \ker(\varepsilon)) = \alpha(x)$ . By Noether isomorphism, we have

$$P_0 / \ker(\varepsilon) \cong M.$$

i.e. there exists a unique  $\hat{\alpha}$  such that

$$\begin{array}{ccc} & & M \\ & \nearrow \hat{\alpha} & \uparrow \varepsilon_x \\ N & \xleftarrow{\hat{\alpha}} & P_0 / \ker(\varepsilon) \end{array}$$

commutes. □

Modules  $\xrightarrow{m \mapsto P_x}$  Projective Resolutions  $\xrightarrow{P_x \rightarrow \text{Hom}(P_x, N)}$  Cochain Complex  $\xrightarrow{H^n}$  Abelian Groups  
Category and Functors.

**Definition 1.26.** A category  $\mathcal{C}$ , the objects is given by  $\text{Obj}(\mathcal{C})$ . For any  $X, Y \in \text{Obj}(\mathcal{C})$ , a set  $\text{Hom}_{\mathcal{C}}(X, Y)$  is a morphism from  $X$  to  $Y$ . For any  $X, Y, Z \in \text{Obj}(\mathcal{C})$ , a composition map

$$\begin{aligned} \text{Hom}_{\mathcal{C}}(Y, Z) \times \text{Hom}_{\mathcal{C}}(X, Y) &\rightarrow \text{Hom}_{\mathcal{C}}(X, Z) \\ (\alpha, \beta) &\mapsto \alpha \circ \beta \end{aligned}$$

Any object  $X$  has  $\text{id}_X \in \text{Hom}(X, X)$  where  $\text{id}_X \circ \alpha = \alpha$  and  $\alpha \circ \text{id}_X = \alpha$ .

**Example.**

1.  $\text{Vect}_{\mathbb{F}}$  Vector space and  $X$  linear map.
2.  $\text{Grp}$ : Groups and group homomorphisms.
3.  $\text{Ab}$ : Abelian groups and group homomorphism.
4.  $\text{TOP}$ : Topological spaces and a smooth map.
5.  $\text{DIFF}$ : smooth manifold with diffeomorphism.
6.  $\Lambda$  ring,  $\text{Mod}_\Lambda$ :  $\Lambda$ -modules.

**Definition 1.27** (Functors). Given that  $\mathcal{C}$  and  $\mathcal{D}$  are two categories. We say that  $F$  is a functor if

$$F : \mathcal{C} \rightarrow \mathcal{D}$$

$$X \mapsto F(X)$$

In addition,

$$X \xrightarrow{\alpha} Y \mapsto F(X) \xrightarrow{F(\alpha)} F(Y)$$

and  $F(\text{id}_X) = \text{id}_{F(X)}$ .

**Definition 1.28.** Covariant is given by  $F(\alpha \circ \beta) = F(\alpha) \circ F(\beta)$ . Contravariant is  $F(\alpha \circ \beta) = F(\beta) \circ F(\alpha)$ .

Ring  $\Lambda$ : Chian( $\Lambda$ ). Objects sequence

$$\cdots \longrightarrow C_{n+1} \xrightarrow{\partial_{n+1}} C_n \xrightarrow{\partial_n} C_{n-1} \longrightarrow \cdots$$

each  $C_i$  is a  $\Lambda$ -module.  $\partial_n : C_n \rightarrow C_{n-1}$  is  $\Lambda$ -linear and  $\partial_n \circ \partial_{n+1} = 0$ . Morphism is

$$\begin{array}{ccccccccccc} C_* = ( & \longrightarrow & C_{n+1} & \xrightarrow{\partial_{n+1}} & C_n & \xrightarrow{\partial_n} & \cdots & \xrightarrow{\partial_2} & C_1 & \xrightarrow{\partial_1} & C_0 & \xrightarrow{\varepsilon} & M & \longrightarrow & 0) \\ & & \downarrow f_{n+1} & & \downarrow f_n & & & & \downarrow f_1 & & \downarrow f_0 & & \downarrow f & & \\ D_* = ( & \longrightarrow & D_{n+1} & \xrightarrow{\delta_{n+1}} & D_n & \xrightarrow{\delta_n} & \cdots & \xrightarrow{\delta_2} & D_1 & \xrightarrow{\delta_1} & D_0 & \xrightarrow{\varepsilon'} & M' & \longrightarrow & 0) \end{array}$$

where  $f = (f_r)_r, F_r : C_r \rightarrow D_r$  and  $f \circ g = (f_r \circ g_r)_r$ .

Homology is a covariant functor.  $H_n : \text{Chain}(\Lambda) \rightarrow \text{Ab}$  where  $H_n(C_*) = \ker \partial_n / \text{Im}(\partial_{n+1})$ .

Cochain complexes: Cochain( $\Lambda$ ) is given by

$$C^* = ( \longrightarrow C^{n-1} \xrightarrow{\delta^n} C^n \xrightarrow{\delta^{n+1}} C^{n+1} \longrightarrow \cdots$$

where  $\delta^n : C^{n-1} \rightarrow C^n$  is  $\Lambda$ -linear and  $\delta^{n+1} \circ \delta^n = 0$  and cohomology is covariant functor:  $H^n(C^*) = \ker \delta^{n+1} / \text{Im}(\delta^n)$ .

Exactness of chain complexes:

$$0 \longrightarrow A_* \xrightarrow{i} B_* \xrightarrow{p} C_* \longrightarrow 0$$

where  $i = (i_n) : A_n \rightarrow B_n$  and  $p = (p_n) : B_n \rightarrow C_n$ . For each  $n$ ,

$$0 \longrightarrow A_n \xrightarrow{i_n} B_n \xrightarrow{p_n} C_n \longrightarrow 0$$

is exact. Likewise, exactness for cochain complex.

**Theorem 1.29** (Algebraic Mayer-Vietoris Theorem).

Let

$$0 \longrightarrow A_* \xrightarrow{i} B_* \xrightarrow{p} C_* \longrightarrow 0$$

be exact sequence of chain complexes. Then there exists a long exact sequence

$$H_{n+1}(B_*) \xrightarrow{p_*} H_{n+1}(C_*) \xrightarrow{\partial} H_n(A_*) \xrightarrow{i_*} H_n(B_*) \xrightarrow{p_*} H_n(C_*) \xrightarrow{\partial} H_{n-1}(A_*)$$

The cochain version

$$0 \longrightarrow A^* \xrightarrow{i^*} B^* \xrightarrow{p^*} C^* \longrightarrow 0$$

Homological algebra has two long exact sequence:

1. Coefficient exact sequence:  $M$  be a  $\Lambda$ -module and

$$0 \longrightarrow A \xrightarrow{i} B \xrightarrow{p} C \longrightarrow 0$$

exact sequence of  $\Lambda$ -modules. Then

$$H^n(M, B) \xrightarrow{p} H^n(M, C) \xrightarrow{\delta} H^n(M, A) \xrightarrow{i} H^{n+1}(M, B) \xrightarrow{p} H^{n+1}(M, C) \xrightarrow{\delta} H^{n+2}(M, A).$$

$M$  fixed, coefficients vary.

2. Reverse Exact sequence"

$$0 \longrightarrow K \longrightarrow L \longrightarrow M \longrightarrow 0$$

be an exact sequence of  $\lambda$ -module.  $N$  fixed coefficient module

$$H^{n-1}(L, N) \xrightarrow{i^*} H^{n-1}(K, N) \xrightarrow{\delta} H^n(M, N) \xrightarrow{p^*} H^n(L, N) \xrightarrow{i^*} H^n(K, N) \cdots$$

$M$  fixed, let

$$0 \longrightarrow A \xrightarrow{i} B \xrightarrow{p} C \longrightarrow 0$$

Let

$$\mathcal{F}_* = \left( \cdots F_{n+1} \xrightarrow{\partial_{n+1}} F_n \xrightarrow{\partial_n} \cdots \xrightarrow{\partial_2} F_1 \xrightarrow{\partial_1} F_0 \xrightarrow{\varepsilon} M \longrightarrow 0 \right)$$

be a fixed free resolutions. Assume each  $F_r$  is finitely generated over  $\Lambda$ . Each  $F_r \cong \Lambda^{r(n)}$ ,  $r(n) \in \mathbb{Z}_{\geq 0}$  is the rank. We will

consistently

$$0 \longrightarrow \text{Hom}(\mathcal{F}, A) \longrightarrow \text{Hom}(\mathcal{F}, B) \longrightarrow \text{Hom}(\mathcal{F}, C) \longrightarrow 0$$

**Proposition 1.30.**

$$\text{Hom}_\Lambda(\Lambda, N) \cong N$$

**Corollary 1.31.** If

$$0 \longrightarrow A \xrightarrow{i} B \xrightarrow{p} C \longrightarrow 0$$

is exact, we have an exact sequence

$$0 \longrightarrow \text{Hom}_\Lambda(\Lambda, A) \longrightarrow \text{Hom}_\Lambda(\Lambda, B) \longrightarrow \text{Hom}_\Lambda(\Lambda, C) \longrightarrow 0$$

**Proposition 1.32.**  $\text{Hom}_\Lambda(\Lambda^m, N) \cong N^{\oplus m}$ .

**Proof.** Let  $(e_i)_{1 \leq i \leq m}$  be a basis for  $\Lambda^m$ . Then

$$\begin{aligned} \text{Hom}_\Lambda(\Lambda^m, N) &\rightarrow N^m \\ \alpha &\mapsto (\alpha(e_1), \dots, \alpha(e_m)) \end{aligned}$$

□

**Corollary 1.33.** If

$$0 \longrightarrow A \xrightarrow{i} B \xrightarrow{p} C \longrightarrow 0$$

is exact then

$$0 \longrightarrow \text{Hom}_\Lambda(\Lambda^n, A) \longrightarrow \text{Hom}_\Lambda(\Lambda^n, B) \longrightarrow \text{Hom}_\Lambda(\Lambda^n, C) \longrightarrow 0$$

is also exact for each  $m \geq 1$ .

$\text{Hom}(\mathcal{F}_*)$  is a cochain complex

$$\text{Hom}(\mathcal{F}_*, N) = (0 \longrightarrow \text{Hom}(F_0, N) \xrightarrow{\partial_1^*} \text{Hom}(F_1, N) \xrightarrow{\partial_2^*} \text{Hom}(F_2, N) \xrightarrow{\partial_3^*} \dots)$$

is a cochain complex.

**Theorem 1.34.** Let

$$\mathcal{F}_* = \left( \dots F_{n+1} \xrightarrow{\partial_{n+1}} F_n \xrightarrow{\partial_n} \dots \xrightarrow{\partial_2} F_1 \xrightarrow{\partial_1} F_0 \xrightarrow{\varepsilon} M \longrightarrow 0 \right)$$

be a free resolution and

$$0 \longrightarrow A \xrightarrow{i} B \xrightarrow{p} C \longrightarrow 0$$

then

$$0 \longrightarrow \text{Hom}(\mathcal{F}_*, A) \xrightarrow{i_*} \text{Hom}(\mathcal{F}_*, B) \xrightarrow{p_*} \text{Hom}(\mathcal{F}_*, C) \longrightarrow 0$$

is an exact sequence of cochain complexes.

$H^n(M, A) := H^n(\text{Hom}(\mathcal{F}_*, A))$ . By algebraic MV Theorem, there exists long exact sequence

$$0 \longrightarrow H^0(M, A) \longrightarrow H^0(M, B) \longrightarrow H^0(M, C) \longrightarrow H^1(M, A) \longrightarrow H^1(M, B) \longrightarrow \dots$$

Eventually we have

$$H^n(M, A) \longrightarrow H^n(M, B) \longrightarrow H^n(M, C) \longrightarrow H^{n+1}(M, A) \longrightarrow H^{n+1}(M, B) \longrightarrow \dots$$

This is equivalent to

$$0 \longrightarrow \text{Hom}(M, A) \xrightarrow{i} \text{Hom}(M, B) \xrightarrow{p} \text{Hom}(M, C) \xrightarrow{\delta} H^1(M, A) \longrightarrow H^1(M, B) \longrightarrow \dots$$

**Proposition 1.35.**  $P \in \text{Mod}_\Lambda$  is projective if and only if  $H^1(P, N) = 0$  for all  $N$ .

**Proof.** If  $P$  is projective, then  $H^1(P, N) = 0$  for all  $N$ . Since we have the following projective resolution for  $P$ :

$$\begin{array}{ccccccc} 0 & \longrightarrow & 0 & \longrightarrow & P & \xrightarrow{\text{id}} & P \longrightarrow 0 \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ P_2 & \longrightarrow & P_1 & \longrightarrow & P_0 & \longrightarrow & M \longrightarrow 0 \end{array}$$

Conversely, if  $H^1(P, N) = 0$  for all  $N$  then  $P$  is projective. Let  $(e_i)_{i \in I}$  be a spanning set for  $P$ . Let  $F_i$  be free with basis  $(E_i)_{i \in I}$ . We have an exact sequence

$$0 \longrightarrow K \xrightarrow{i} F \xrightarrow{\pi} P \longrightarrow 0$$

$\pi(E_i) = e_i$  and  $\pi$  surjective. Put  $K \subset \ker(\pi)$ . Take  $H^*(P, \cdot)$  with coefficients  $m$ . Then

$$0 \longrightarrow H^0(P, K) \longrightarrow H^0(P, F) \longrightarrow H^0(P, P) \longrightarrow \dots$$

i.e.

$$0 \longrightarrow \text{Hom}(P, K) \xrightarrow{i} \text{Hom}(P, K) \xrightarrow{p} \text{Hom}(P, P) \longrightarrow 0$$

exact.  $p : \text{Hom}(P, F) \rightarrow \text{Hom}(P, P)$  is surjective. Choose an  $s \in \text{Hom}(P, F)$  such that  $p^*(s) = \text{id}_P$  so  $p \circ s = \text{id}$  so  $s$  splits the exact sequence on the right so  $s$  splits the exact sequence

$$0 \longrightarrow K \xrightarrow{i} F \xrightarrow{p} P \longrightarrow 0$$

so sequence splits and  $F \cong P \oplus K$  so  $P$  is projective.  $\square$

**Lemma 1.36** (Schanuel's lemma). Let

$$0 \longrightarrow I \xrightarrow{i} P \xrightarrow{p} M \longrightarrow 0$$

$$0 \longrightarrow J \xrightarrow{j} Q \xrightarrow{q} M \longrightarrow 0$$

be exact where  $P, Q$  are projective then

$$I \oplus Q \cong J \oplus P.$$

Define  $P \times Q \in \text{Mod}_\Lambda$  by

$$P \times Q = \{(x, y) \in P \times Q, p(x) = q(y)\}$$

we have the following commutative diagram

$$\begin{array}{ccc} P \times Q & \xrightarrow{\pi_2} & Q \\ \downarrow \pi_1 & & \downarrow q \\ P & \xrightarrow{p} & M \end{array}$$

We get two exact sequence

$$0 \longrightarrow \ker(\pi_1) \longrightarrow P \times Q \xrightarrow{\pi_1} P \longrightarrow 0$$

$$0 \longrightarrow \ker(\pi_2) \longrightarrow P \times Q \xrightarrow{\pi_2} P \longrightarrow 0$$

Both  $\pi_1, \pi_2$  are surjective. Let  $x \in P$  need to produce  $y \in Q$  such that  $q(y) = p(x)$ . But  $q : Q \rightarrow M$  is surjective,  $\ker(\pi_1) = \{(x, y) \in P \times Q \mid p(x) = q(y), x = 0\} = \{(0, y) \in P \times Q \mid q(y) = 0\} \cong \ker(q) = J$ . So we have the following exact sequence s

$$0 \longrightarrow J \longrightarrow P \times Q \longrightarrow P \longrightarrow 0$$

$$0 \longrightarrow I \longrightarrow P \times Q \longrightarrow Q \longrightarrow 0$$

$P$  and  $Q$  projective so sequences split. Thus,

$$Q \oplus I \cong P \times Q \cong P \oplus J.$$

We then have for  $k \geq 1$ :

$$H^k(I, N) \oplus H^k(P', N) \cong H^k(J, N) \oplus H^k(P, N).$$

so

$$H^k(I, N) \cong H^k(J, N).$$

which is known as syzygy.

## 1.5 30/01/2026

**Theorem 1.37.** Let

$$0 \longrightarrow A \xrightarrow{i} B \xrightarrow{p} C \longrightarrow 0$$

be an exact sequence of  $\Lambda$ -module and let  $(P_* \rightarrow A)$  and  $(Q_* \rightarrow C)$  be projective resolution. Then there exists projective resolution  $T_* \rightarrow B$  such that

1.  $T_n = P_n \oplus Q_n$  for each  $n$ .
2.  $\partial_n^T : T_n \rightarrow T_{n-1}$  has the form of the upper triangular matrix with diagonal  $\partial_n^P$  to  $\partial_n^Q$ .

**Lemma 1.38.** Suppose given commutative diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & D & \xrightarrow{\hat{i}} & E & \xrightarrow{\hat{p}} & F \longrightarrow 0 \\ & & \downarrow \alpha & & \downarrow \beta & & \downarrow \gamma \\ 0 & \longrightarrow & A & \xrightarrow{i} & B & \xrightarrow{p} & C \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ & & 0 & & 0 & & 0 \end{array}$$

we have exact sequence such that  $\alpha, \beta, \gamma$  are surjective. Then

1.  $\hat{p} : \ker(\beta) \rightarrow \ker(\gamma)$ .
2.  $\hat{p}$  is surjective.
3.  $\hat{i} : \ker(\alpha) \rightarrow \ker(\beta)$ .
4.  $\ker(\hat{p}|_{\ker(\beta)}) = \text{Im}(\hat{i})|_{\ker(\alpha)}$ .

**Proof.**

1. Let  $x \in \ker(\beta)$  then  $\beta(x) = 0$ . So  $p \circ \beta(x) = 0$  which implies  $\gamma \circ \hat{p}(x) = 0$  i.e.  $\hat{p}(x) \in \ker(\gamma)$ .
2. Let  $y \in \ker(\gamma)$ . As  $\hat{p}$  is surjective, then there exists  $x' \in E$  such that  $\hat{p}(x') = y$ . Thus,  $\gamma \circ \hat{p}(x') = 0$  i.e.  $p \circ \beta(x') = 0$ . Hence,  $\beta(x') \in \ker(p) = \text{Im}(i)$ . So there exists  $a \in A$  such that  $i(a) = \beta(x')$ . Since  $\alpha$  is surjective, choose  $d \in D$  such that  $\alpha(d) = a$ . Then

$$i \circ \alpha(d) = \beta(x') \implies \beta \hat{i}(d) = \beta(x')$$

this implies  $x' - \hat{i}(d) \in \ker(\beta)$ . Put  $x = x' - \hat{i}(d)$ . We have  $\hat{p}(x) = \hat{p}(x') = y$  since  $\hat{p} \circ \hat{i} = 0$ .

3. Let  $z \in \ker(\alpha)$  i.e.  $i \circ \alpha(z) = 0$  so  $p \circ \hat{i}(z) = 0$ . Hence  $\hat{i}(z) \in \ker(\beta)$ .
4. Let  $x \in \ker(\beta)$  be such that  $\hat{p}(x) = 0$ . Hence  $p \circ \beta(x) = 0$  i.e.  $j \circ \hat{p}(x) = 0$  where  $j$  is the inclusion  $j : \ker(\gamma) \hookrightarrow C$ ,  $\beta(x) \in \text{Im}(i)$ . But  $j$  is injective this implies  $\hat{p}(x) = 0$ . There exists a  $d \in D$  such that  $\hat{i}(d) = x$ .  $\beta \circ \hat{i}(d) = i \circ \alpha(d) = \beta(x) = 0$ .  $\alpha(d) = 0$  so  $d \in \ker(\alpha)$  there exists some  $d \in \ker(\alpha)$  such that  $\hat{i}(d) = x$ .

□

**Proof of the Theorem 1.37.** Now, suppose given exact sequence of  $\Lambda$ -module

$$0 \longrightarrow A \xrightarrow{i} B \xrightarrow{p} C \longrightarrow 0$$

and let  $P_* \rightarrow A$ ,  $Q_* \rightarrow C$  be projective resolutions. Consider

$$\begin{array}{ccccccc} 0 & \longrightarrow & P_0 & \xrightarrow{\hat{i}} & P_0 \oplus Q_0 & \xrightarrow{\pi} & Q_0 \longrightarrow 0 \\ & & \downarrow \varepsilon & & & & \downarrow \eta \\ 0 & \longrightarrow & A & \xrightarrow{i} & B & \xrightarrow{p} & C \longrightarrow 0 \\ & & \downarrow & & & & \downarrow \\ & & 0 & & & & 0 \end{array}$$

We want to construct some map from  $P_0 \oplus Q_0 \rightarrow B$ . By projectivity

$$\begin{array}{ccc} & P_0 \oplus Q_0 & \\ \mu \swarrow & \downarrow \eta \circ \pi & \\ B & \xrightarrow{p} & C \end{array}$$

Since  $P_0 \oplus Q_0$  is projective, there exists some  $\mu : P_0 \oplus Q_0 \rightarrow B$  such that  $p \circ \mu = \eta \circ \pi$ . Then  $\mu$  looks like

$$\mu = \begin{pmatrix} \varepsilon & ? \\ 0 & \eta \end{pmatrix}$$

Now consider the diagram

$$\begin{array}{ccccccc} & & 0 & & 0 & & 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & \ker(\varepsilon) & \longrightarrow & \ker(\mu) & \longrightarrow & \ker(\eta) \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \eta \\ 0 & \longrightarrow & P_0 & \xrightarrow{\hat{i}} & P_0 \oplus Q_0 & \xrightarrow{\pi} & Q_0 \longrightarrow 0 \\ & & \downarrow \varepsilon & & \downarrow \mu & & \downarrow \eta \\ 0 & \longrightarrow & A & \xrightarrow{i} & B & \xrightarrow{p} & C \longrightarrow 0 \end{array}$$

Now repeat with  $A' = \ker(\varepsilon)$ ,  $B' = \ker(\mu)$  and  $C' = \ker(\eta)$ . So is  $P_* \rightarrow A$ ,  $Q_* \rightarrow C$ . Notice

$$\begin{array}{ccccc}
 P_1 & \longrightarrow & P_1 \oplus Q_1 & \longrightarrow & Q_1 \\
 \downarrow & & \downarrow & & \downarrow \\
 \ker(\varepsilon) & \longrightarrow & \ker(\mu) & \longrightarrow & \ker(\eta) \\
 \downarrow & & \downarrow & & \downarrow \\
 P_0 & \longrightarrow & P_0 \oplus Q_0 & \longrightarrow & Q_0 \\
 \downarrow \varepsilon & & \downarrow & & \downarrow \eta \\
 A & & B & & C \\
 \downarrow & & \downarrow & & \downarrow \\
 0 & & 0 & & 0
 \end{array}$$

We can define  $T_n = P_n \oplus Q_n$ . Then we have

$$\begin{array}{ccccccc}
 0 & \longrightarrow & P_2 & \longrightarrow & T_2 & \longrightarrow & Q_2 \longrightarrow 0 \\
 & & \downarrow \partial_2^P & & \downarrow \partial_2^T & & \downarrow \partial_2^Q \\
 0 & \longrightarrow & P_1 & \longrightarrow & T_1 & \longrightarrow & Q_1 \longrightarrow 0 \\
 & & \downarrow \partial_1^P & & \downarrow \partial_1^T & & \downarrow \partial_1^Q \\
 0 & \longrightarrow & P_0 & \longrightarrow & T_0 & \longrightarrow & Q_0 \longrightarrow 0 \\
 & & \downarrow \varepsilon & & \downarrow \mu & & \downarrow \eta \\
 0 & \longrightarrow & A & \xrightarrow{i} & B & \xrightarrow{p} & C \longrightarrow 0 \\
 & & \downarrow & & \downarrow & & \downarrow \\
 & & 0 & & 0 & & 0
 \end{array}$$

□

So given exact sequence

$$0 \longrightarrow A \xrightarrow{i} B \xrightarrow{p} C \longrightarrow 0$$

and projective resolutions  $P_* \rightarrow A$  and  $Q_* \rightarrow C$ . There exists projective resolution  $T_* \rightarrow B$  where  $T_n = P_n \oplus Q_n$  and an exact sequence of chain complexes

$$\begin{array}{ccccccc}
 0 & \longrightarrow & P_* & \xrightarrow{\hat{i}} & T_* & \xrightarrow{\hat{p}} & Q_* \longrightarrow 0 \\
 & & \downarrow & & \downarrow & & \downarrow \\
 0 & \longrightarrow & A & \xrightarrow{i} & B & \xrightarrow{p} & C \longrightarrow 0
 \end{array}$$

To get reverse exact sequence we apply  $\text{Hom}(\cdot, N)$ .

## 1.6 02/02/2026

We want to investigate the form of  $\partial_n^T$ . In every row, we have the trivial exact sequence

$$0 \longrightarrow P_n \longrightarrow P_n \oplus Q_n \longrightarrow Q_n \longrightarrow 0$$

So we claim that

$$\partial_n^T = \begin{pmatrix} \partial_n^P & ? \\ 0 & \partial_n^Q \end{pmatrix}$$

Write

$$\partial_n^T = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

where  $a : P_n \rightarrow P_{n-1}$ ,  $b : Q_n \rightarrow P_{n-1}$ ,  $c : P_n \rightarrow Q_{n-1}$ ,  $d : Q_n \rightarrow Q_{n-1}$ . We know

$$\pi_{n-1} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \partial_n^Q \circ \pi_n$$

Let  $\begin{pmatrix} x \\ y \end{pmatrix} \in P_n \oplus Q_n$  then

$$\pi_{n-1} \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \partial_n \circ \pi_n \begin{pmatrix} x \\ y \end{pmatrix}$$

i.e.

$$\pi_{n-1} \begin{pmatrix} ax + by \\ cx + dy \end{pmatrix} = cx + dy = \partial_n^Q(y)$$

This implies  $c = 0$  and  $d = \partial_n^Q$ . Similarly,

$$\begin{pmatrix} a & b \\ 0 & \partial_n^Q \end{pmatrix} \circ i = i \circ \partial_n^P$$

i.e.

$$ax = \partial_n^P x \implies a = \partial_n^P$$

**Theorem 1.39.** Let

$$0 \longrightarrow A \xrightarrow{i} B \xrightarrow{p} C \longrightarrow 0$$

be an exact sequence of  $\Lambda$ -module. Let  $P_* \rightarrow A$  and  $Q_* \rightarrow C$  be projective resolutions. Then there exists projective resolutions  $T_* \rightarrow B$  such that  $T_n = P_n \oplus Q_n$  for all  $n$  and

$$\partial_n^T = \begin{pmatrix} \partial_n^P & ? \\ 0 & \partial_n^Q \end{pmatrix}$$

And for each  $n$

$$0 \longrightarrow P_n \longrightarrow P_n \oplus Q_n \longrightarrow Q_n \longrightarrow 0$$

is given by

$$x \mapsto \begin{pmatrix} x \\ 0 \end{pmatrix}, \quad \begin{pmatrix} x \\ y \end{pmatrix} \mapsto y$$

**Lemma 1.40.** Let

$$0 \longrightarrow P \xrightarrow{i} P \oplus Q \xrightarrow{\pi} Q \longrightarrow 0$$

be a split exact sequence. Then for any  $N$ , the sequence

$$0 \longrightarrow \text{Hom}(Q, N) \xrightarrow{\pi^*} \text{Hom}(P \oplus Q, N) \xrightarrow{i^*} \text{Hom}(P, N) \longrightarrow 0$$

is also split exact.

**Proof.** Let  $s : Q \rightarrow P \oplus Q$  be a right splitting, and  $r : P \oplus Q \rightarrow Q$  be a left splitting i.e.  $\pi \circ s = \text{id}_Q$  and  $r \circ i = \text{id}_P$ . Then

$$s^* \circ \pi^* = \text{id}_{\text{Hom}(Q, N)} \text{ and } i^* \circ r^* = \text{id}_{\text{Hom}(P, N)}$$

□

Reverse Exact Sequence: Let

$$0 \longrightarrow A \xrightarrow{i} B \xrightarrow{p} C \longrightarrow 0$$

be an exact sequence of  $\Lambda$ -module. For any coefficient module  $N$ , we get a long exact sequence of cohomology.

$$0 \longrightarrow H^0(C, N) \xrightarrow{p^*} H^0(B, N) \xrightarrow{i^*} H^0(A, N) \xrightarrow{\delta} H^1(C, N) \xrightarrow{p^*} H^1(B, N) \xrightarrow{i^*} H^1(A, N) \xrightarrow{\delta} \dots$$

**Proof.** Take the exact sequence of projective resolutions

$$\begin{array}{ccccccc} 0 & \longrightarrow & P_* & \longrightarrow & T_* & \longrightarrow & Q_* \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & A & \longrightarrow & B & \longrightarrow & C \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ & & 0 & & 0 & & 0 \end{array}$$

Apply  $\text{Hom}(\cdot, N)$ . Then as we have seen the sequence of cochain complex we see

$$0 \longrightarrow \text{Hom}(Q_*, N) \xrightarrow{\pi^*} \text{Hom}(T_*, N) \xrightarrow{i^*} \text{Hom}(P_*, N) \longrightarrow 0$$

is exact by applying MV Theorem.

□

Start by considering short exact sequence of  $\Lambda$ -modules:

$$\mathcal{X} = (0 \longrightarrow K \xrightarrow{i} X \xrightarrow{p} M \longrightarrow 0)$$

If

$$\mathcal{Y} = (0 \longrightarrow K' \xrightarrow{i'} Y \xrightarrow{p'} M' \longrightarrow 0)$$

is also exact by a morphism  $x \rightarrow y$  meaning a commutative diagram of  $\Lambda$ -modules.

$$\begin{array}{ccccccc} 0 & \longrightarrow & K & \xrightarrow{i} & X & \xrightarrow{p} & M \longrightarrow 0 \\ & & \downarrow f_K & & \downarrow f & & \downarrow f_M \\ 0 & \longrightarrow & K' & \xrightarrow{i'} & Y & \xrightarrow{p'} & M' \longrightarrow 0 \end{array}$$

Write  $\mathcal{X} \equiv \mathcal{X}'$  (congruent) when there exists such commutative diagram. Both congruence and isomorphisms are equivalence relations.

**Remark.** In particular,

$$\begin{array}{ccccccc} 0 & \longrightarrow & K & \xrightarrow{i} & X & \xrightarrow{p} & M \longrightarrow 0 \\ & & \downarrow \text{id} & & \downarrow f & & \downarrow \text{id} \\ 0 & \longrightarrow & K & \xrightarrow{i} & X & \xrightarrow{p} & M \longrightarrow 0 \end{array}$$

By Five Lemma,  $X \cong X'$ .

Fix  $M, N$ , let  $\mathcal{E}xt(M, N) =$  all exact sequence :

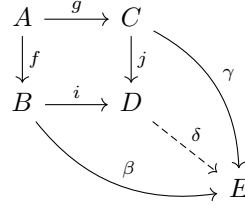
$$0 \longrightarrow N \longrightarrow ? \longrightarrow M \longrightarrow 0$$

And  $\text{Ext}(M, N) := \mathcal{E}xt(M, N)/\equiv$  the set of congruence class of such extension.

**Theorem 1.41** (Baer's Theorem).

1.  $\text{Ext}(M, N)$  is "naturally" an additive abelian group.
2.  $\text{Ext}(M, N) \cong H^1(M, N)$ .

Push-outs and Pull-backs. Let  $\mathcal{C}$  be some category and let



$(f, g)$  have a "push-out"  $D$  with some object in  $\mathcal{C}$  when there exists  $i : B \rightarrow D$  and  $j : C \rightarrow D$  such that given any morphisms  $\beta : B \rightarrow E$ ,  $\gamma : C \rightarrow E$  such that  $\beta \circ f = \gamma \circ g$ . There exists unique map  $\delta : D \rightarrow E$  making the diagram commutes.

**Proposition 1.42.**  $\text{Mod}_\Lambda$  has push-outs.

**Proof.** Given

$$\begin{array}{ccc} A & \xrightarrow{g} & C \\ \downarrow f & & \\ B & & \end{array}$$

Define  $\underline{\lim}(f, g) = (B \oplus C)/\text{Im}(-f \times g)$ . Define

$$\begin{aligned} i : B &\rightarrow \underline{\lim}(f, g) \\ j : C &\rightarrow \underline{\lim}(f, g) \end{aligned}$$

where

$$i(b) = \begin{pmatrix} b \\ 0 \end{pmatrix}, \quad j(c) = \begin{pmatrix} 0 \\ c \end{pmatrix}$$

Claim  $\underline{\lim}(f, g)$  is a push-out of  $f, g$ . So suppose we have  $\Lambda$ -homomorphisms

$$\beta : B \rightarrow E, \quad \gamma : C \rightarrow E$$

such that  $\beta \circ f = \gamma \circ g$ . There exists morphism  $(\beta, \gamma) : B \oplus C \rightarrow E$  via

$$(\beta, \gamma) \begin{pmatrix} x \\ y \end{pmatrix} = \beta(x) + \gamma(y).$$

Suppose  $\begin{pmatrix} x \\ y \end{pmatrix} \in \text{Im}(-f \times g)$ . So there exists  $z \in A$  such that

$$\begin{pmatrix} -f(z) \\ g(z) \end{pmatrix} \in B \oplus C$$

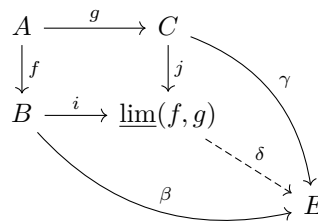
$$(\beta, \gamma) \begin{pmatrix} -f(z) \\ g(z) \end{pmatrix} = -\beta f(z) + \gamma g(z) = 0$$

by commutativity.  $(\beta, \gamma)\text{Im}(-f \times g) \equiv 0$ . SO  $(\beta, \gamma)$  induces a unique homomorphism

$$\delta = (\beta, \gamma) : \underline{\lim}(f, g) \rightarrow E$$

making the diagram commutes. □

The dual notation is "pull-back". We have the push-out diagram



The pull-back diagram looks like

$$\begin{array}{ccc}
 E & & \\
 \searrow & & \searrow \\
 & \lim_{\leftarrow}(f, g) & \longrightarrow C \\
 & \downarrow & \downarrow g \\
 & B & \xrightarrow{f} A \\
 & \downarrow & \\
 & C & \\
 & \downarrow g & \\
 & B & \xrightarrow{f} A
 \end{array}$$

has a pull-back  $\lim_{\leftarrow}(f, g)$  when there exists  $\lim_{\leftarrow}(f, g) \xrightarrow{\pi_B} B$  and  $\lim_{\leftarrow}(f, g) \xrightarrow{\pi_C} C$  such that given  $\beta : E \rightarrow B$ ,  $\gamma : E \rightarrow C$  such that  $f \circ \beta = g \circ \gamma$ . There exists unique  $\delta : E \rightarrow \lim_{\leftarrow}(f, g)$  making diagram commutes.

**Proposition 1.43.**  $\text{Mod}_\Lambda$  has pull-backs.

**Proof.** Given

$$\begin{array}{ccc}
 & C & \\
 & \downarrow g & \\
 B & \xrightarrow{f} & A
 \end{array}$$

Define

$$\lim_{\leftarrow}(f, g) = \left\{ \begin{pmatrix} x \\ y \end{pmatrix} \in B \times C : f(x) = g(y) \right\}$$

then  $\lim_{\leftarrow}$  is subject of  $B \times C$ . □

Given an exact sequence in  $\text{Mod}_\Lambda$ :

$$0 \longrightarrow K \xrightarrow{i} P \xrightarrow{p} M \longrightarrow 0$$

and a homomorphism  $f : K \rightarrow N$ . Claim: there exists exact sequence

$$0 \longrightarrow N \xrightarrow{j} \lim_{\rightarrow}(f, i) \xrightarrow{\pi} M \longrightarrow 0$$

such that the following commutes

$$\begin{array}{ccccccc}
 0 & \longrightarrow & K & \xrightarrow{i} & P & \xrightarrow{p} & M \longrightarrow 0 \\
 & & \downarrow f & & \downarrow \eta & & \downarrow \text{id} \\
 0 & \longrightarrow & N & \xrightarrow{j} & \lim_{\rightarrow}(f, i) & \xrightarrow{\pi} & M \longrightarrow 0
 \end{array}$$

**Proof.**

$$\begin{array}{ccc}
 K & \xrightarrow{i} & P \\
 \downarrow f & & \downarrow \eta \\
 N & \xrightarrow{j} & \lim_{\rightarrow}(f, i)
 \end{array}$$

where  $j(n) = \begin{pmatrix} n \\ 0 \end{pmatrix}$ ,  $\eta(x) = \begin{pmatrix} 0 \\ x \end{pmatrix}$ . Need to construct  $\pi : \lim_{\rightarrow}(f, i) \rightarrow M$ . Construction:

$$\begin{array}{ccc}
 K & \xrightarrow{i} & P \\
 \downarrow f & & \downarrow \eta \\
 N & \xrightarrow{j} & \lim_{\rightarrow}(f, i)
 \end{array}
 \begin{array}{c}
 \searrow p \\
 \nearrow \pi \\
 \searrow 0
 \end{array}
 \begin{array}{c}
 \\
 \\
 M
 \end{array}$$

$p \circ i = 0 = 0 \circ f$ .  $\pi$  is the mapping guaranteed the push-out property. Observe that diagram commutes.

$$\pi \begin{pmatrix} n \\ x \end{pmatrix} = p(x)$$

Need to show  $\ker(\pi) = \text{Im}(j)$ . Let  $\begin{pmatrix} n \\ x \end{pmatrix} \in \lim_{\rightarrow}(f, i)$  then

$$\pi \begin{pmatrix} n \\ x \end{pmatrix} = p(x)$$

i.e.

$$\pi \begin{pmatrix} n \\ x \end{pmatrix} = 0 \Leftrightarrow p(x) = 0 \Leftrightarrow x = i(k) \text{ for some } k \in K$$

But

$$\ker(\pi) = \left\{ \begin{pmatrix} n \\ i(k) \end{pmatrix} = \begin{pmatrix} n \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ i(k) \end{pmatrix} \right\}$$

But since

$$\begin{pmatrix} 0 \\ i(k) \end{pmatrix} = \begin{pmatrix} -f(k) \\ 0 \end{pmatrix}$$

We see that  $n - f(k) \in \text{Im}(j)$  i.e.  $\ker(\pi) = \text{Im}(j)$ . □

**Definition 1.44.** Given an exact sequence

$$\mathcal{P} = (0 \longrightarrow K \xrightarrow{i} P \xrightarrow{p} M \longrightarrow 0)$$

and a morphism  $f : K \rightarrow N$  we get an exact sequence

$$f_*(\mathcal{P}) = (0 \longrightarrow N \longrightarrow \varinjlim(f, i) \longrightarrow M \longrightarrow 0)$$

Main step: If  $\mathcal{P}$  is projective then any element of  $\text{Ext}(M, N)$  is obtaining in this way.

## 1.7 06/02/2026

The non-obvious part of Baer's Theorem is  $\text{Ext}(M, N)$  has natural structure of an abelian group. The useful point is  $\text{Ext}(M, N) \cong H^1(M, N)$ .

Pick  $M$ , choose an exact sequence of the following form

$$\mathcal{P} = (0 \longrightarrow K \xrightarrow{i} P \xrightarrow{p} M \longrightarrow 0)$$

where  $P$  is projective. We will produce a map

$$\begin{aligned} \underline{\text{lim}} : \text{Hom}(K, M) &\rightarrow \text{Ext}(M, N) \\ f &\mapsto f_*(\mathcal{P}) \end{aligned}$$

where  $f_*(\mathcal{P})$  is the following exact sequence

$$f_*(\mathcal{P}) = (0 \longrightarrow N \xrightarrow{j} \varinjlim(f, i) \xrightarrow{\pi} M \longrightarrow 0)$$

We have to define  $j, \pi$  and we need to show that  $\pi$  is surjective and  $j$  is injective.  $\ker(\pi) = \text{Im}(j)$ ,  $\nu, j$  are from the universal property of push-out.

$$\begin{array}{ccccc} K & \xrightarrow{i} & P & & \\ \downarrow f & & \downarrow \nu & \searrow p & \\ N & \xrightarrow{j} & (N \oplus P)/\text{Im}(-f \times i) & \xrightarrow{\pi} & M \\ & \searrow 0 & & & \uparrow \\ & & & & \end{array}$$

To define  $\pi : N \oplus P/\text{Im}(-f \times i) \rightarrow M$ , observe we have a mapping

$$N \oplus P \rightarrow M$$

$$\begin{pmatrix} n \\ x \end{pmatrix} \mapsto p(x)$$

$$\begin{pmatrix} -f(n) \\ i(y) \end{pmatrix} \mapsto 0 = p \circ i(y).$$

surjective. So we get  $\pi : N \oplus P/\text{Im}(-f \times i) \rightarrow M$  well-defined surjective map. Clear that  $\pi \circ j = 0$ ,  $j(n) = \begin{pmatrix} n \\ 0 \end{pmatrix} \mapsto 0$ . Since  $\ker(j) \subset \ker(\pi)$ . Let  $\begin{pmatrix} n \\ x \end{pmatrix} \in \ker(\pi)$ .

$$\pi \begin{pmatrix} n \\ x \end{pmatrix} = p(x) = 0$$

$x \in \ker(p) = \text{Im}(i)$ . Write  $x = i(y)$  we have

$$\begin{pmatrix} n \\ i(y) \end{pmatrix} = \begin{pmatrix} n - f(y) \\ 0 \end{pmatrix} \in \text{Im}(j)$$

so  $f_*(\mathcal{P})$  is well-defined.

**Definition 1.45.** By a projective 1-stem, I mean exact sequence

$$\mathcal{P} = (0 \longrightarrow K \xrightarrow{i} P \xrightarrow{p} M \longrightarrow 0)$$

where  $P$  is projective.

**Theorem 1.46.** Let

$$\mathcal{P} = (0 \longrightarrow K \xrightarrow{i} P \xrightarrow{p} M \longrightarrow 0)$$

be a projective 1-stem: where the mapping  $\underline{\lim} : \text{Hom}(K, N) \rightarrow \text{Ext}(M, N)$ ,  $f \mapsto f_*(\mathcal{P})$  i.e. every extension

$$0 \longrightarrow N \longrightarrow X \longrightarrow M \longrightarrow 0$$

is congruent to  $f_*(\mathcal{P})$  for some  $f$ .

**Proof.** Let

$$\mathcal{E} = (0 \longrightarrow N \xrightarrow{\hat{i}} X \xrightarrow{\hat{p}} M \longrightarrow 0)$$

and

$$\begin{array}{ccccccc} \mathcal{P} = (0 & \longrightarrow & K & \xrightarrow{i} & P & \xrightarrow{p} & M \longrightarrow 0 \\ & & \downarrow f & & \downarrow \text{id} & & \downarrow \text{id} \\ 0 & \longrightarrow & N & \longrightarrow & X & \xrightarrow{\hat{p}} & M \end{array}$$

$p : P \rightarrow M$  by definition is the canonical map. There exists  $\hat{\text{id}} : P \rightarrow X$  such that  $\hat{p} \circ \hat{\text{id}} = p$  so  $\hat{p} \circ \hat{\text{id}} \circ i = p \circ i = 0$  i.e.  $\hat{\text{id}}(\text{Im}(i)) \subset \ker(p)$ . Given the choice of  $\hat{\text{id}}$ , there exists unique  $f : K \rightarrow N$  making the diagram commute. Now compare  $f_*(\mathcal{P})$  with  $\mathcal{E}$ :

$$\begin{array}{ccccccc} 0 & \longrightarrow & K & \xrightarrow{i} & P & \longrightarrow & M \longrightarrow 0 \\ & & \downarrow f & & \downarrow \nu & & \downarrow \text{id} \\ 0 & \longrightarrow & N & \xrightarrow{j} & \underline{\lim}(f, i) & \longrightarrow & M \longrightarrow 0 \\ & & \downarrow & & \downarrow \mu & & \downarrow \text{id} \\ 0 & \longrightarrow & N & \longrightarrow & X & \longrightarrow & M \longrightarrow 0 \end{array}$$

$j, \nu$  exists from the universal property.  $\mu : \underline{\lim}(f, i) \rightarrow X$  exists since

$$\begin{array}{ccc} K & \xrightarrow{i} & P \\ \downarrow f & & \downarrow \nu \\ N & \xrightarrow{j} & \underline{\lim}(f, i) \end{array} \quad \begin{array}{c} \searrow \hat{\text{id}} \circ i \\ \downarrow \mu \\ X \end{array}$$

so we have a congruence

$$f_*(\mathcal{P}) \xrightarrow{\mu} \mathcal{E}$$

□

So we have a surjective mapping

$$\text{Hom}(P, N) \xrightarrow{i^*} \text{Hom}(K, N) \xrightarrow{\underline{\lim}} \text{Ext}(M, N)$$

**Proposition 1.47.** If  $P \rightarrow N$  so  $i^*(g) = g \circ i : K \rightarrow N$ . Then  $\underline{\lim}(i^*(g))$  has the trivial extension.

**Proof.** Consider

$$\begin{array}{ccccccc} \mathcal{P} = (0 & \longrightarrow & K & \xrightarrow{i} & P & \longrightarrow & M \longrightarrow 0 \\ & & \downarrow i^*(g) & \nearrow g & \downarrow & & \downarrow \\ 0 & \longrightarrow & N & \longrightarrow & \underline{\lim}(i^*(g), i) & \longrightarrow & M \longrightarrow 0 \end{array}$$

Claim there exists left lifting of the bottom row. Define  $r : \underline{\lim}(i^*(g), i) \rightarrow N$  by

$$r\left(\begin{smallmatrix} n \\ x \end{smallmatrix}\right) = n + g(x)$$

$r \circ j(n) = r\left(\begin{smallmatrix} n \\ 0 \end{smallmatrix}\right) = n$ ,  $r$  is a left splitting.

□

**Proposition 1.48.** Let  $(A, +, 0)$  be additive abelian group. Suppose  $(B, *)$  is a magma, no assumption on  $*$  but  $* : B \times B \rightarrow B$ . Suppose there exists a surjective  $\varphi : A \rightarrow B$ , and  $\varphi(a_1 + a_2) = \varphi(a_1) * \varphi(a_2)$ . Then  $(B, *, \varphi(0))$  is an abelian group. And

$$\varphi : (A, +, 0) \rightarrow (B, *, \varphi(0))$$

is a surjective group homomorphism.  $\ker(\varphi) = \varphi^{-1}(\varphi(0))$ . In particular,

$$(B, *, \varphi(0)) \cong (A, +, 0) / \ker(\varphi).$$

**Proof.**  $(*)$  is associative. Choose  $b_1, b_2, b_3 \in B$ , we need to show

$$(b_1 * b_2) * b_3 = b_1 * (b_2 * b_3)$$

Since  $\varphi$  is surjective, choose  $a_i \in A$  such that  $\varphi(a_i) = b_i$ . Then

$$(b_1 * b_2) * b_3 = (\varphi(a_1) * \varphi(a_2)) * \varphi(a_3) = \varphi(a_1 + a_2) * \varphi(a_3) = \varphi((a_1 + a_2) + a_3).$$

But  $(A, +)$  is associative, it follows that

$$\varphi((a_1 + a_2) + a_3) = \varphi(a_1 + (a_2 + a_3)) = b_1 * (b_2 * b_3).$$

Identity. We claim that  $\varphi(0)$  is the identity of  $(B, *)$ . Choose  $b \in B$  then

$$b * \varphi(0) = \varphi(a) * \varphi(0) \text{ for some } a \in A.$$

which is equivalent to  $\varphi(a) = b$ . Likewise,  $\varphi(0) * b = b$ .

Inverses. Let  $b \in B$ , choose  $a \in A$  such that  $\varphi(a) = b$ . Put  $b' = \varphi(-a)$  then

$$b * b' = \varphi(a) * \varphi(-a) = \varphi(0)$$

Similarly,  $b' * b = \varphi(0)$ . So  $(B, *, \varphi(0))$  is a group, in particular, an abelian group.  $\square$

## 1.8 09/02/2026

If  $M, N$  are  $\Lambda$ -modules, we need to show

1.  $\text{Ext}(M, N)$  has natural structure of an additive abelian group.

2.  $\text{Ext}(M, N) \cong H_\Lambda^1(M, N)$ .

Strategy of proof:

$$0 \longrightarrow K \xrightarrow{i} P \xrightarrow{p} M \longrightarrow 0$$

be a projective, 1-stem. For any  $N$ , there exists an exact sequence

$$0 \longrightarrow \text{Hom}_\Lambda(M, N) \xrightarrow{p^*} \text{Hom}_\Lambda(P, N) \xrightarrow{i^*} \text{Hom}_\Lambda(K, N) \longrightarrow H^1(M, N) \longrightarrow 0$$

We also have  $\underline{\text{lim}} : \text{Hom}(K, N) \rightarrow \text{Ext}(M, N)$ ,  $\underline{\text{lim}}(f) = f_*(\mathcal{P})$ . We know  $\underline{\text{lim}}$  is surjective. We need to construct  $*$  :  $\text{Ext}(M, N) \times \text{Ext}(M, N) \rightarrow \text{Ext}(M, N)$  such that

$$\underline{\text{lim}}(f + g) = \underline{\text{lim}}(f) * \underline{\text{lim}}(g).$$

If I can do this, then I know  $\text{Ext}(M, N)$  is an abelian group. In particular,

$$\text{Ext}(M, N) = \text{Hom}_\Lambda(M, N) / \underline{\text{lim}}^{-1}(\underline{\text{lim}}(0)).$$

How do we define  $+$  on  $\text{Hom}(K, N)$ ?

$$+ : \text{Hom}(K, N) \times \text{Hom}(K, N) \rightarrow \text{Hom}(K, N)$$

$$(f + g)(x) \mapsto f(x) + g(x)$$

$$\text{Hom}(K \times K, N \times N) \xrightarrow{\Delta^*} \text{Hom}(K, N \times N)$$

$$\times \uparrow$$

$$\downarrow \text{Add}_*$$

$$\text{Hom}(K, N) \times \text{Hom}(K, N) \longrightarrow \text{Hom}(K, N)$$

where  $(f \times g)(m_1, m_2) = (f(m_1), g(m_2))$ .  $\Delta : K \rightarrow K \times K$  is given by  $\Delta(x) = (x, x)$  so that  $\Delta^* : \text{Hom}(K \times K, N \times N) \rightarrow \text{Hom}(K, N \times N)$ ,  $\Delta^*(F) = F \circ \Delta = F(x, x)$  so that

$$\Delta^*(f \times g) = (f(x), g(x))$$

And

$$\text{Add}_* : \text{Hom}(K, N \times N) \rightarrow \text{Hom}(K, N)$$

$$(F_1(x), F_2(x)) \mapsto F_1(x) + F_2(x)$$

So  $(\text{Add}_* \circ \Delta^* \circ \times)(f, g) = f + g$ .  $x$  is well-defined by

$$x_1 = (0 \longrightarrow N \xrightarrow{i_1} x_1 \xrightarrow{p_1} M \longrightarrow 0)$$

$$x_2 = (0 \longrightarrow N \xrightarrow{i_2} x_2 \xrightarrow{p_2} M \longrightarrow 0)$$

then

$$x_1 \times x_2 = (0 \longrightarrow N \times N \xrightarrow{i_1 \times i_2} x_1 \times x_2 \xrightarrow{p_1 \times p_2} M \times M \longrightarrow 0)$$

To define  $\Delta^*$  we recall "pull-back" of an extension. Consider "Schanuel's Lemma":

$$\alpha^*(x) = (0 \longrightarrow K \longrightarrow X \times M' \xrightarrow{\pi_{M'}} M' \longrightarrow 0)$$

$$\downarrow \pi_x$$

$$\downarrow \alpha$$

$$x = (0 \longrightarrow K \xrightarrow{i} X \xrightarrow{p} M \longrightarrow 0)$$

Here,  $X \times M' = \{(x, m') \in X \times M', p(x) = \alpha(m')\}$ .  $\pi_x(x, m') = x$  and  $\pi_{M'}(x, m') = m'$ . Thus,  $\alpha^*(x)$  is the pull-back. In addition,

$$\Delta^* : \text{Ext}(M \times M, N \times N) \rightarrow \text{Ext}(M, N \times N)$$

$$x_1 \times x_2 \mapsto \Delta^*(x_1 \times x_2).$$

$Add : N \times N \rightarrow N$ , this is a  $\Lambda$ -homomorphism.

$$(Add_*)(x) = \varinjlim (Add)$$

We need to check the following diagram commutes

$$\begin{array}{ccccc}
 & & \text{Ext}(M \times M, N \times N) & \xrightarrow{\quad} & \text{Ext}(M, N \times N) \\
 & \nearrow \varinjlim & \uparrow \Delta^* & & \nearrow \varinjlim \\
 \text{Hom}(K \times K, N \times N) & \xrightarrow{\quad} & \text{Hom}(K, N \times N) & & \\
 \uparrow \times & & \downarrow Add_* & & \downarrow \\
 & \nearrow \varinjlim \times \varinjlim & \text{Ext}(M, N) \times \text{Ext}(M, N) & \xrightarrow{\quad} & \text{Ext}(M, N) \\
 \text{Hom}(K, N) \times \text{Hom}(K, N) & \xrightarrow{\quad + \quad} & \text{Hom}(K, N) & & \\
 & & \nearrow \varinjlim & & 
 \end{array}$$

Last time we proved that  $\varinjlim$  is the trivial extension.

$$\begin{array}{ccccccc}
 \text{Hom}(P, N) & \xrightarrow{i^*} & \text{Hom}(K, N) & \xrightarrow{\varinjlim} & \text{Ext}(M, N) & \longrightarrow & 0 \\
 \downarrow \text{id} & & \downarrow \text{id} & & \downarrow & & \\
 \text{Hom}(P, N) & \xrightarrow{i^*} & \text{Hom}(K, N) & \xrightarrow{\delta} & H^1(M, N) & \longrightarrow & 0
 \end{array}$$

We have isomorphism due to Noether.

**Example.** Let  $M = N = \mathbb{Z}/3$ . Compute  $H^1(\mathbb{Z}/2, \mathbb{Z}/3)$ . Take a free resolution for  $\mathbb{Z}/3$  over  $\mathbb{Z}$

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\times 3} \mathbb{Z} \longrightarrow \mathbb{Z}/3 \longrightarrow 0$$

To compute  $H^k(\mathbb{Z}/3, \mathbb{Z}/2)$ , compute

$$0 \longrightarrow \text{Hom}_{\mathbb{Z}}(\mathbb{Z}, \mathbb{Z}/3) \xrightarrow{\times 3} \text{Hom}_{\mathbb{Z}}(\mathbb{Z}, \mathbb{Z}/3) \longrightarrow 0$$

i.e.

$$0 \longrightarrow \mathbb{Z}/3 \xrightarrow{0} \mathbb{Z}/3 \longrightarrow 0$$

Hence,  $H^0(\mathbb{Z}/3, \mathbb{Z}/3) \cong H^1(\mathbb{Z}/3, \mathbb{Z}/3) \cong \mathbb{Z}/3$ . Baer's Theorem says there are exact three congruence chain of extensions

$$0 \longrightarrow \mathbb{Z}/3 \longrightarrow ? \longrightarrow \mathbb{Z} \longrightarrow 0$$

corresponding to  $\{0, 1, 2\}$ . Taking 0, we have the trivial extension. 1, 2 both give  $\mathbb{Z}/9$ , but in two non-congruent ways. For 1, we have

$$0 \longrightarrow \mathbb{Z}/3 \xrightarrow{i} \mathbb{Z}/9 \xrightarrow{p} \mathbb{Z}/3 \longrightarrow 0$$

where  $i(0) = 0, i(1) = 3, i(2) = 6$ .  $p : x \mapsto [x]_3$ . For 2, it's more complicated via the map  $\alpha([x]_9) = [2x]_9$ .

**Example.** Let  $\Lambda = \mathbb{Z}$ ,  $M = \mathbb{Z}/3$  and  $N = \mathbb{Z}/4$ . Consider

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\times 3} \mathbb{Z} \longrightarrow \mathbb{Z}/3 \longrightarrow 0$$

Applying  $\text{Hom}(\cdot, \mathbb{Z}/4)$  we get

$$0 \longrightarrow \mathbb{Z}/4 \xrightarrow{\times 3} \mathbb{Z}/4 \longrightarrow 0$$

$\times 3 : \mathbb{Z}/4 \rightarrow \mathbb{Z}/4$  is an isomorphism so  $H^1(\mathbb{Z}/3, \mathbb{Z}/4) = 0$ . So there exists a unique

$$0 \longrightarrow \mathbb{Z}/4 \longrightarrow \mathbb{Z}/4 \oplus \mathbb{Z}/3 \longrightarrow \mathbb{Z}/3 \longrightarrow 0$$

Cohomological dimension, projective dimension. Fix a ring  $\Lambda$  and a module  $M$ .

**Definition 1.49** (Cohomological dimension).  $cd(M) \leq n$  if and only if  $H^{n+1}(M, N) = 0$  for all  $N$ . So

$$cd(M) := \min\{n : \forall N, H^{n+1}(M, N) = 0\}$$

If there's no such  $n$ , then  $cd(M) = \infty$ .

**Definition 1.50** (Projective dimension).  $pd(M) \leq n$  if and only if there exists projective resolution of length  $n$

$$0 \longrightarrow P_n \longrightarrow P_{n-1} \longrightarrow \cdots \longrightarrow P_0 \longrightarrow M \longrightarrow 0$$

If there's no such  $n$ , write  $pd(M) = \infty$ .

**Proposition 1.51.** If  $pd(M) \leq n$ , then  $cd(M) \leq n$ .

**Proof.** Let

$$0 \longrightarrow P_n \xrightarrow{\partial_n} P_{n-1} \xrightarrow{\partial_{n-1}} \cdots \xrightarrow{\partial_1} P_0 \xrightarrow{\varepsilon} M \longrightarrow 0$$

be a resolution with each  $P_i$  projective (maybe zero). Take  $N$  and applying  $\text{Hom}_\Lambda(\cdot, N)$  we get

$$0 \longrightarrow \text{Hom}(P_0, N) \longrightarrow \cdots \longrightarrow \text{Hom}(P_n, N) \longrightarrow 0$$

Using this projective resolution, we see that  $H^{n+1}(M, N) = 0$  for all  $N$ . This is independent choice of  $P$ . So  $pd(M) \leq n$  implies  $cd(M) \leq n$ .  $\square$

Dimension shifting. Let

$$P_{n+1} \xrightarrow{\partial_{n+1}} P_n \longrightarrow \cdots \longrightarrow P_2 \xrightarrow{\partial_2} P_1 \xrightarrow{\partial_1} P_0 \xrightarrow{\varepsilon} M \longrightarrow 0$$

Consider  $K = \ker(\varepsilon) = \text{Im}(\partial_1)$ , we now have a projective resolution of  $K$ .

$$P_{n+1} \xrightarrow{\partial_{n+1}} P_n \longrightarrow \cdots \longrightarrow P_2 \xrightarrow{\partial_2} P_1 \xrightarrow{\partial_1} K \longrightarrow 0$$

Replace  $Q_4 = P_{r+1}$ ,  $\delta_r = \partial_{r+1}$  to get

$$Q_n \xrightarrow{\delta_n} Q_{n-1} \longrightarrow \cdots \longrightarrow Q_0 \longrightarrow K \longrightarrow 0$$

Compute  $H^*(M, N)$  and  $H^*(K, N)$  we get

$$H^n(K, N) \cong H^{n+1}(M, N)$$

called the dimension shifting. We could repeat by defining  $L = \ker(\partial_1)$  to get

$$H^{n+2}(M, N) \cong H^n(L, N).$$

**Proposition 1.52.**  $cd(M) \leq n$  implies  $pd(M) \leq n$ . In particular, we have  $cd(M) = pd(M)$ .

**Proof.** Let

$$\longrightarrow P_n \longrightarrow P_{n-1} \longrightarrow \cdots \longrightarrow P_1 \xrightarrow{\partial_1} P_0 \longrightarrow M \longrightarrow 0$$

Let  $K = \ker(\partial_{n-1})$ , we get an exact sequence

$$0 \longrightarrow K \longrightarrow P_{n-1} \longrightarrow \cdots \longrightarrow P_0 \longrightarrow M \longrightarrow 0$$

where by dimension shifting

$$H^{n+1}(M, N) \cong H^1(K, N)$$

Now suppose that  $cd(M) \leq n$  then  $H^{n+1}(M, N) = 0$  for all  $N$ . So  $H^1(K, N) = 0$  for all  $N$  i.e.  $K$  is projective. Thus,

$$0 \longrightarrow K \longrightarrow P_{n-1} \longrightarrow \cdots \longrightarrow P_0 \longrightarrow M \longrightarrow 0$$

is a projective resolution of length  $\leq n$  i.e.  $pd(M) \leq n$ .  $\square$

**Definition 1.53.** Let  $\Lambda$  be a ring,  $gd(\Lambda) \leq n$  if and only if  $cd(M) \leq n$  for all  $n$  if and only if  $H^{n+1}(M, N) = 0$  for all  $M$  and  $N$ .  $gd(n)$  is called global dimension of  $\Lambda$ .

**Example.**  $gd(\mathbb{F}) = 0$ .  $gd(\mathbb{F}_1 \times \mathbb{F}_2 \times \cdots \times \mathbb{F}_n) = 0$ .  $gd(\mathbb{Z}) = 1$ . If  $A$  is a P.I.D not a field,  $gd(A) = 1$ .

**Example.**  $gd(\mathbb{Z}(G)) = \infty$ ,  $gd(\mathbb{F}(G)) = 0$  if  $\text{char}(\mathbb{F}) \nmid |G|$ .

**Theorem 1.54 (Hilbert Syzygy Theorem).** If  $A$  is a commutative ring, and  $A[t]$  be a polynomial ring of one variable. Then  $gd(A(t)) = gd(A) + 1$ . In particular

$$gd(A[t_1, \dots, t_n]) = gd(A) + n.$$

## 1.9 13/02/2026

**Definition 1.55.** Let  $\Lambda$  be a ring and  $M \in \text{Mod}_\Lambda$ .  $M$  is Noetherian when given a sequence of submodules  $(M_r)_{r \in \mathbb{N}}$  with  $M_r \subset M_{r+1}$  then there exists  $N$  such that  $M_r = M_N$  for all  $r \geq N$ .

**Proposition 1.56.** If  $M$  is Noetherian then  $M$  is finitely generated over  $\Lambda$ .

**Proof.** If  $\{e_1, \dots, e_n\} \subset M$  and

$$[e_1, \dots, e_n] = \left\{ \sum_{i=1}^n e_i \lambda_i \mid \lambda_i \in \Lambda \right\}$$

Clearly  $[e_1, \dots, e_n]$  is a submodule of  $M$ . Choose  $e_1 \in \subset$ , if  $M = [e_1]$  we are done. Otherwise, choose  $e_2 \in M$  such that  $e_2 \notin [e_1]$ . If  $[e_1, e_2] = M$ , we are done. Suppose  $\{e_1, \dots, e_n\} \in M$  while  $[e_1, \dots, e_n] \neq M$ , choose  $e_{n+1} \in M \setminus [e_1, \dots, e_n]$  so that  $[e_1, \dots, e_n] \subsetneq [e_1, \dots, e_{n+1}]$ . If we can continue this process independently we obtain a sequence  $(M_r)_{r \in \mathbb{N}}$ ,  $M_r \subsetneq M_{r+1}$ , a contradiction to Noetherian condition so there exists  $N$  such that  $M =$

$[e_1, \dots, e_N]$ . □

**Proposition 1.57.** If  $K \subset M$  is a  $\Lambda$ -submodule, and  $M$  is Noetherian then  $K$  is Noetherian.

**Proof.** A submodule of  $K$  is a submodule of  $M$ . □

**Proposition 1.58.** Let  $P : M \rightarrow Q$  be a surjective  $\Lambda$ -homomorphism. If  $M$  is Noetherian then  $Q$  is Noetherian.

**Proof.** Let  $(Q_r)_{r \in \mathbb{N}}$  be an ascending sequence of submodules in  $Q$  with  $(Q_r \subset Q_{r+1})$ . Then  $(P^{-1}(Q_r))_{r \in \mathbb{N}}$  is an ascending sequence in  $M$  so there exists  $N$  such that  $P^{-1}(Q_r) = P^{-1}(Q_N)$  for all  $r \geq N$ .  $Q_r = P(P^{-1}(Q_r))$  so that  $Q_r = Q_N$  for all  $r \geq N$  as required. □

**Proposition 1.59.** Let

$$0 \longrightarrow K \xrightarrow{i} M \xrightarrow{p} Q \longrightarrow 0$$

be an exact sequence of  $\Lambda$ -modules. If  $K$  and  $Q$  are finitely generated over  $\Lambda$  then  $M$  is finitely generated over  $\Lambda$ .

**Proof.** Let  $(e_r)_{1 \leq r \leq q}$  be a finitely generated set for  $Q$ . Let  $(\varphi_s)_{1 \leq s \leq k}$  be a finitely generated set for  $K$ . As  $p$  is surjective, choose  $E_r \in M$  such that  $p(E_r) = e_r$ . Claim that

$$\{E_r\}_{1 \leq r \leq q} \cup \{\varphi_s\}_{1 \leq s \leq k}$$

is a generating set of  $M$ . Taking  $m \in M$  write  $p(m) = \sum_{r=1}^q e_r \lambda_r \in Q$ . Put  $m' = m - \sum_{r=1}^q E_r \lambda_r$  then  $p(m) = p(m')$  so  $m - m' \in \ker(p) = K$ . Write  $m - m' = \sum_{s=1}^k q_s H_s$ ,  $H_s \in \Lambda$ . Then

$$m = \sum_{r=1}^q e_r \lambda_r + \sum_{s=1}^k q_s H_s$$

as desired. □

**Theorem 1.60.** Let

$$0 \longrightarrow K \xrightarrow{i} M \xrightarrow{p} Q \longrightarrow 0$$

be an exact sequence of  $\Lambda$ -modules. Then  $M$  is Noetherian if and only if  $K$  and  $Q$  are Noetherian.

**Proof.** ( $\implies$ ) : See above.

( $\impliedby$ ) : Let  $(M_r)_{r \in \mathbb{N}}$  be an ascending sequence of submodules of  $M$ . Put  $M' = \bigcup_{r \geq 0} M_r$ ,  $M'$  is a submodule of  $M$ . We need to show that there exists some  $N$  such that  $M_N = M'$ . Put  $Q' = p(M')$ ,  $K' = M' \cap K$ . Now we have an exact sequence

$$0 \longrightarrow K' \longrightarrow M' \longrightarrow Q'$$

$Q$  Noetherian implies  $Q'$  Noetherian and  $K$  implies  $K'$  Noetherian.  $K$  and  $Q$  Noetherian implies both  $K'$  and  $Q'$  are finitely generated. So  $M'$  is finitely generated. Let  $(E_r)_{1 \leq r \leq N}$  be a finitely generating set for  $M$ . For each  $t$ , there exists  $\alpha(t)$ , such that  $E_r \in E_{\alpha(t)}$ . So  $[E_1, \dots, E_N] = M'$  and  $[E_1, \dots, E_N] \subset \bigcup_{t=1}^N M_{\alpha(t)}$ . Put  $T = \max\{\alpha(t)\}$ ,  $M' \subset M_T$ .  $M_r = M'$  for all  $r \geq T$  as required. □

**Definition 1.61.** Let  $R$  be a commutative ring. Say that  $R$  is Noetherian ring when any ascending sequence of ideals terminates i.e.  $I_r \subset I_{r+1}$  then there exists  $N$  such that  $I_r = I_N$  for all  $r \geq N$ . If  $R$  is a Noetherian ring  $R$  then  $R$  is Noetherian as a module over itself.

**Proposition 1.62.**  $R$  is Noetherian if and only if every finitely generating  $R$ -modules is Noetherian.

**Proof.** ( $\implies$ ) : If  $R$  is Noetherian then each  $\underbrace{R \oplus \dots \oplus R}_n$  is a Noetherian  $R$ -module i.e. every free generated free

$R$ -module is Noetherian. Let  $M$  be a finitely generating  $R$ -module  $\{e_1, \dots, e_n\}$  finitely generating set. Let  $R^n$  be the free  $R$ -module with bases  $\{E_1, \dots, E_n\}$ . Consider  $p : R^n \rightarrow M$  surjective,  $p(E_i) = e_i$ .  $p$  is surjective so  $M$  is Noetherian.

( $\impliedby$ ) : Trivial since  $R$  is generated by a single element. □

**Definition 1.63.**  $R$  is right Noetherian when ascending sequence of right terminates. Same definition for left Noetherian.

## 1.10 23/02/2026

**Theorem 1.64** (Hilbert Basis Theorem). Let  $R$  be a commutative Noetherian ring. Then  $R[t]$  is also Noetherian.

**Proof.** Let  $I \subset R[t]$  be an ideal. It is enough to show that  $I$  is finitely generated. Let  $\alpha(t) \in R[t]$ , and

$$\alpha(t) = a_0 + a_1t + \cdots + a_nt^n \quad a_n \neq 0$$

The "initial term" in  $\text{in}(\alpha)$  is  $a_n$  which is the coefficient of the highest non-zero term. Define  $J \subset R$  by

$$J = \{\text{in}(\alpha_1)\lambda_1 + \cdots + \text{in}(\alpha_m)\lambda_m, \quad \lambda_i \in R, \alpha_i \in I\}$$

$J$  is a submodule of  $R$ . We prove by comparing  $I$  and  $J$ . Hypothesis on  $R$  is that  $R$  is Noetherian, in particular,  $J$  is finitely generated over  $R$ . We shall "construct" a finite set of generators as follows. Choose  $\alpha_1(t) \in I \setminus \{0\}$  of lowest possible degree, if  $(\text{in}(\alpha_1)) = J$  we are done. If not, choose  $\alpha_2(t) \in I$  of lowest possible degree such that  $\text{in}(\alpha_2) \notin J_1$ . Now look at  $(\text{in}(\alpha_1), \text{in}(\alpha_2))$ , if  $(\text{in}(\alpha_1), \text{in}(\alpha_2)) = J$  we are done. Otherwise, we repeat. So if  $\alpha_1(t), \dots, \alpha_k(t)$  are chosen so that  $\deg(\alpha_1) \leq \cdots \leq \deg(\alpha_k)$ . But  $(\text{in}(\alpha_1), \dots, \text{in}(\alpha_k)) \neq J$ . Next step is to choose  $\alpha_{k+1}$  such that  $\text{in}(\alpha_{k+1}) \notin J_k$ . We are constructing an ascending sequence

$$J_1 = (\text{in}(\alpha_1)) \subset J_2 = (\text{in}(\alpha_1), \text{in}(\alpha_2)) \subset \cdots$$

As  $R$  is Noetherian, this process would eventually terminate. Then  $J = (\text{in}(\alpha_1), \dots, \text{in}(\alpha_n))$ . We claim that  $I = (\alpha_1, \dots, \alpha_n)$ . Suppose not, put  $I_n = (\alpha_1, \dots, \alpha_n)$ . Then there exists  $g(t) \in I$  such that  $g(t) \notin I_n$ . Choose  $g(t) \in I$  of the lowest possible degree such that  $g(t) \notin I_n$ , write

$$g(t) = b_0 + b_1t + b_2t^2 + \cdots + b_Nt^N$$

By definition of  $J$ ,  $b_n \in J = (\text{in}(\alpha_1), \dots, \text{in}(\alpha_n))$ . Thus,  $b_n$  is a linearly combinations of  $\text{in}(\alpha_i)$ . Observe  $\deg(\alpha_i) \leq \deg(g)$  for all  $i$ , by previous choices.

$$\alpha_i(t) = \text{in}(\alpha_i)t^{e_i} + \text{terms of lower degree}$$

$e_i \leq N$  for all  $i$ . Write

$$\tilde{\alpha}_i(t) = \alpha_i t^{N-e_i}$$

so that  $\deg(\tilde{\alpha}_i(t)) = \deg(g) = N$ . Put

$$\tilde{g}(t) = \sum_{i=1}^n \tilde{\alpha}_i(t)\lambda_i$$

Now  $g(t)$  and  $\tilde{g}(t)$  have the same highest terms. Thus,  $\deg(g(t) - \tilde{g}(t)) < N$ . But  $\tilde{g}(t) \in I_n$  while  $g(t) \notin I_n$ . Therefore,  $g(t) - \tilde{g}(t) \notin I_n$  and has lower degree than  $g$ , a contradiction by the choice of  $g$ . Therefore,  $I = I_n = (\alpha_1, \dots, \alpha_n)$ . Hence  $R[t]$  is Noetherian.  $\square$

**Definition 1.65.** Let  $R$  be a commutative ring,  $R$  is a principal ideal ring if every ideal in  $R$  is generated by a single element. In particular,  $R$  is an integral domain (no-zero divisor).

**Examples.**

- (i)  $\mathbb{Z}$  is a P.I.D.
- (ii) If  $\mathbb{F}$  is a field,  $\mathbb{F}[t]$  is a P.I.D.

Both rely on the division algorithm.

**Proof.**

- (i) Obviously,  $\mathbb{Z}$  is an integral domain. Let  $I \subset \mathbb{Z}$  be a proper ideal, let  $d \in I$  be the smallest possible positive element. Let  $a \in I$ , suppose  $a \geq 0$  then  $d \leq a$  and  $a = dq + r$  so  $r \in I$  which means  $r = 0$ . Thus,  $I = (d)$ .
- (ii) If  $\mathbb{F}$  is a field,  $\mathbb{F}[t]$  is a P.I.D. which follows easily by the degree of polynomial.  $\square$

**Proposition 1.66.** If  $R$  is a principal ideal ring, then  $R$  is a Noetherian/

**Proof.** Every ideal is generated by a single element.  $\square$

**Corollary 1.67.** If  $R$  is a principal ideal ring then  $R[t]$  is Noetherian.

**Corollary 1.68.** If  $R$  is a principal ideal ring, then  $R[t_1, \dots, t_n]$  is Noetherian.

We shall classify all finitely generated modules

**Theorem 1.69.**

- (i) If  $M \in \text{Mod}_R$  then

$$M \cong F \oplus \Theta \quad F \text{ free}, \Theta \text{ torsion}$$

For each prime  $p$ ,  $\Theta_p = p$ -torsion of  $\Theta$  and

- (ii)  $\Theta = \Theta_{p_1} + \cdots + \Theta_{p_n}$ .
- (iii) JNF, classifying  $\Theta_p$ .

Let  $R$  be a commutative integral domain. If  $M \in \text{Mod}_R$ , define  $\Theta(M) \subset M$  by

$$\Theta(M) = \{m \in M \mid \exists \lambda \in R, \lambda \neq 0, m\lambda = 0\}$$

**Proposition 1.70.**  $\Theta(M)$  is a submodule of  $M$ .

**Proof.** If  $0 \in \Theta(M)$ . Let  $m_1, m_2 \in \Theta(M)$ . Choose  $\lambda_1, \lambda_2 \in R, m_1\lambda_1 = m_2\lambda_2 = 0$  where  $\lambda_1, \lambda_2 \neq 0$ . Then

$$(m_1 + m_2)\lambda_1\lambda_2 = m_1\lambda_1\lambda_2 + m_2\lambda_1\lambda_2 \in \Theta(M).$$

If  $m \in \Theta(M)$ ,  $m\lambda = 0$ ,  $\lambda \neq 0$ . If  $\mu \in R$ ,  $(m\mu)\lambda = (m\lambda)\mu = 0$  so that  $\lambda \neq 0$ . Thus, closed under scalar multiplication as well.  $\square$

$\Theta(M)$  is the torsion submodule of  $M$ .

**Definition 1.71.**  $M$  is torsion free when  $\Theta(M) = 0$ .

**Proposition 1.72.**  $\Theta(M/\Theta(M)) = 0$  i.e.  $M/\Theta(M)$  is torsion free.

**Proof.** Let

$$\begin{aligned} \square : M &\rightarrow M/\Theta(M) \\ m &\mapsto m + \Theta(M) \end{aligned}$$

Let  $[m] \in M/\Theta(M)$  i.e. there exists  $\lambda \neq 0$  such that  $[m]\lambda = 0$  i.e.  $m\lambda \in \Theta(M)$ . So there exists  $\mu \in R$  non-zero, such that  $(m\lambda) \cdot \mu = 0$ . So  $m \cdot (\lambda\mu) = 0$ ,  $\lambda\mu \neq 0$ . Thus,  $[m]\lambda = 0$  implies  $\lambda \neq 0$  so  $[m] = 0$ .  $\square$

We get an exact sequence

$$0 \longrightarrow \Theta(M) \longrightarrow M \longrightarrow M/\Theta(M) \longrightarrow 0$$

**Proposition 1.73.** Let  $R$  be a commutative integral domain. Let  $M$  be a torsion free finitely generated  $R$ -module. Then there exists an injective  $R$ -homomorphism  $i : M \rightarrow R^N$  free module of rank  $N$  for some  $N$ . Thus,  $M$  is a submodule of a free module.

**Proof.** Let  $\{m_1, \dots, m_N\}$  be a finitely generated set for  $M$  (over  $R$ ). Let  $\{e_1, \dots, e_m\} \subset \{m_1, \dots, m_N\}$  be a maximal linear independent subset i.e.

$$\sum_{i=1}^n e_i \lambda_i = 0 \implies \lambda_i = 0, \forall i$$

Put  $F = \text{Span}_R\{e_1, \dots, e_n\}$  then  $F$  is a free module of rank  $n$  over  $R$ . Write  $\{f_1, \dots, f_m\} = \{m_1, \dots, m_N\} \setminus \{e_1, \dots, e_n\}$ . For each  $i$ , by maximality of  $\{e_1, \dots, e_n\}$  with linear independence. There exists dependence relation of form

$$f_1 \alpha_i + e_1 \mu_1 + \dots + e_n \mu_n = 0$$

where at least one of  $\alpha_i, \mu_j \neq 0$ . But if  $\alpha_i = 0$ , we have contradiction to the linearly independence of  $\{e_1, \dots, e_n\}$ . Thus, each  $\alpha_i$  is non-zero. Hence,

$$f_i \alpha_i = \sum e_r (\text{some terms}) \in F$$

Put  $\alpha = \alpha_1 \dots \alpha_n$ ,  $f_i \alpha \in F$  for all  $i$ . As  $F\alpha \subset F$ , then  $M\alpha \subset F$ . And we have a homomorphism  $h : M \rightarrow F$ ,  $h(m) = m\alpha$  with  $\alpha$  non-zero. Since  $M$  is torsion free, so  $h(m) = 0$  implies  $m\alpha = 0$  implies  $m = 0$  i.e.  $h$  is injective. Thus,  $M \subset F$ .  $\square$

**Proposition 1.74.** Let  $R$  be a commutative P.I.D. If  $M \in \text{Mod}_R$  finitely generated over  $R$ . If  $M$  embeds as a submodule of a free module, then  $M$  is free.

**Proof.** Suppose  $M \subset R^n$ , we prove by induction on  $n$ . If  $n = 1$ ,  $M$  is generated by an element  $e$ . Suppose true for  $n - 1$ , taking  $R^n$  and decompose to  $R^n = R^{n-1} \oplus R$ . So we have an exact sequence

$$0 \longrightarrow R^{n-1} \xrightarrow{i} R^n \xrightarrow{\pi} R \longrightarrow 0$$

Consider  $\pi(M) \subset R$ . If  $\pi(M) = 0$  then  $M \subset \ker(\pi) \cong R^{n-1}$  so by induction  $M$  is free. If  $\pi(M) \neq 0$ , then we have an exact sequence

$$0 \longrightarrow M \cap R^{n-1} \longrightarrow M \longrightarrow \pi(M) \longrightarrow 0$$

So  $\pi(M)$  is free,  $\pi(M) \cong R$ . Since  $\pi(M)$  is free, the sequence splits. Thus

$$M \cong (M \cap R^{n-1}) \oplus \pi(M) \cong (M \cap R^{n-1}) \oplus R$$

By induction  $M \cap R^{n-1}$  is free, so that  $M \cap R^{n-1} \cong R^m$  so  $M \cong R^m \oplus R \cong R^{m+1}$  free.  $\square$

If we take the last two propositions:

**Theorem 1.75.** Let  $M$  be a finitely generated torsion free module over a commutative P.I.D  $R$ . Then  $M$  is free.

**Proof.**  $M$  torsion free, so  $M \hookrightarrow R^n$  for some  $n$ .  $R$  is a P.I.D implies  $M \cong R^m$  for some  $m$ .  $\square$

**Proposition 1.76.** Let  $M$  be a free module over commutative P.I.D. Then  $M \cong \Theta \oplus F$  where  $\Theta$  is a torsion module and  $F$  is a torsion free module.

**Proof.** We have a short exact sequence

$$0 \longrightarrow \Theta(M) \longrightarrow M \longrightarrow M/\Theta(M) \longrightarrow 0$$

But  $M/\Theta(M)$  is torsion free so  $M/\Theta(M)$  is free by the previous proposition. Then we have the short exact sequence

$$0 \longrightarrow \Theta(M) \longrightarrow M \longrightarrow F \longrightarrow 0$$

Since  $F$  is free, we know that the sequence splits. Thus,  $M \cong \Theta(M) \oplus F$ .  $\square$

**Remark.** Let  $R$  be commutative integral domain,  $M \subset R^M$  torsion free. In general  $M$  is a long way from being free. But if  $M$  is P.I.D. then  $R$  is free. In general  $M \subset P$  with  $P$  projective, it's not necessarily true that  $M$  is projective. Rings with the property  $M \subset P$  projective implies  $M$  projective is called hereditary ring.

**Remark.** Let  $R$  be Noetherian. If  $M \subset F^n$  and  $M$  free then  $M \cong F^m$ .

## 1.11 27/02/2026

We want to classify finitely generated  $R$ -module:

Step 1, For  $M$  finitely generated  $R$ -module then there exists exact sequence

$$0 \longrightarrow \Theta(M) \longrightarrow M \longrightarrow F \longrightarrow 0$$

where  $F$  is free and  $\Theta(M)$  is torsion i.e. for all  $m \in \Theta$ , there exists  $\lambda \in R$  and  $\lambda \neq 0$  such that  $m\lambda = 0$ . As  $F$  is free then sequence splits i.e.

$$M \cong \Theta(M) \oplus F$$

so we want to classify torsion modules. Suppose  $R$  be commutative Noetherian integral domain and  $\Theta$  is a finitely generated torsion module. If  $a \in R$  and  $M \in \text{Mod}_R$  then define

$$\Theta_a(M) = \{m \in M : ma = 0\}$$

**Proposition 1.77.**  $\Theta_a(M) \subset \Theta(M)$ .

**Proof.** Obvious.  $\square$

**Proposition 1.78.** If  $M$  is finitely generated over  $R$ , then there exists  $a \in R$ ,  $a \neq 0$  such that  $\Theta(M) = \Theta_a(M)$ .

**Proof.** Let  $m_1, \dots, m_n$  be a finite generating set for  $\Theta(M)$  since  $R$  is Noetherian. For each  $i$ , choose  $a_i \in R$  with  $a_i \neq 0$  such that  $m_i a_i = 0$ . Take  $a = a_1 \cdots a_n$  then for each  $i$ ,  $m_i a = 0$  while  $a \neq 0$  since  $R$  is an integral domain. Therefore, if  $m \in \Theta(M)$ ,  $m = \sum_{i=1}^n m_i \lambda_i$  then  $ma = 0$  i.e.  $m \in \Theta_a(M)$ .  $\square$

Recall that if  $R$  is a commutative ring,  $a, b, c \in R$

$$(b, c) = \{b\lambda + c\mu \mid \lambda, \mu \in R\}$$

$(b, c)$  is a submodule of  $R$ .  $(b, c) = (1)$  when  $(b, c) = R$ .

**Proposition 1.79.** If  $a = bc$ ,  $(b, c) = (1)$  then  $\Theta_a(M) = \Theta_b(M) + \Theta_c(M)$  (Internal direct sum).

**Proof.** If  $a = bc$ . If  $mb = 0$  or  $mc = 0$  then  $ma = 0$ . Thus,  $\Theta_b(M) + \Theta_c(M) \subset \Theta_a(M)$ . Since  $(b, c) = (1)$ ,  $b\lambda + c\mu = 1$ . Let  $x \in \Theta_a(M)$ ,

$$x = xc\mu + xb\lambda$$

$(xc\mu)b = x(bc)\mu = xa\mu = 0$ . Thus,  $xc\mu \in \Theta_b(M)$ . Similarly,  $xb\lambda \in \Theta_c(M)$ . Therefore,  $\Theta_a(M) \subset \Theta_b(M) + \Theta_c(M)$ . It remains to show that  $\Theta_b(M) \cap \Theta_c(M) = 0$ . If  $x \in \Theta_b(M) \cap \Theta_c(M)$ .  $x = xb\lambda + xc\mu = 0$ .  $\square$

We need to know that if  $a \in R$ , then  $a$  decomposes as a product

$$a = p_1^{e_1} \cdots p_n^{e_n}$$

where each  $p_i$  is prime,  $e_1 \geq 1$ . That is, if  $(p_i, p_j) = 1$  then  $(p_i^{e_i}, p_j^{e_j}) = 1$ . It then follows

$$\Theta_a(M) = \Theta_{p_1^{e_1}}(M) + \cdots + \Theta_{p_n^{e_n}}(M) \cong \Theta_{p_1^{e_1}}(M) \oplus \cdots \oplus \Theta_{p_n^{e_n}}(M).$$

Final Step: Let  $R$  be commutative P.I.D. Let  $p$  be prime

**Theorem 1.80.** Suppose  $\Theta(M) = \Theta_{p^e}(M)$ ,  $p$  prime,  $e \geq 1$ . Then there exists collection of positive integer  $e_1, \dots, e_m$  with  $e_1 \leq \dots \leq e_m$  such that

$$\Theta_{p^e}(M) \cong \bigoplus_{i=1}^m R/p^{e_i}$$

We can then get a complete classification of finitely generated modules/P.I.D. namely

$$M = R^m \oplus \bigoplus R/p_i^{e_i}$$

where  $p_1, \dots, p_n$  are primes and  $e_i \geq 1$ .  $R^m$  is free so  $cd(R^m) = 0$  but we have an exact sequence

$$0 \longrightarrow R \xrightarrow{p^e} R \longrightarrow R/p^e \longrightarrow 0$$

which means  $cd_R(R/p^e) = 1$ . Hence  $cd(M) \leq 1$ . Thus, if  $R$  is commutative P.I.D. then  $gd(R) = 1$ .

## 1.12 02/03/2026

Let  $R$  be a commutative integral domain. We say  $a$  is irreducible when  $a = bc$  implies  $b \in R^*$  or  $c \in R^*$ .  $I$  is prime if and only if  $R/I$  is an integral domain. If  $p \in R$ ,  $p$  is prime if and only if  $R/(p)$  is an integral domain.

**Proposition 1.81.** Let  $R$  be commutative Noetherian integral domain. If  $a \in R$  then there exists irreducible  $a' \in R$  such that  $a = a'b$  for some  $b$ .

**Proof.** Let  $a \in R \setminus \{0\}$ , if  $a$  is irreducible we are done. If not,  $a = a_0b_0$  for some  $a_0, b_0 \in R \setminus R^*$ . If  $a_0$  is irreducible we are done. Otherwise, if  $a_0$  is reducible, write  $a_0 = a_1b_1$ ,  $(a_0) \subset (a_1)$ . If  $a_1$  is irreducible we are done. And we repeat this process. If  $a_n$  is reducible and  $a_n = a_{n+1}b_{n+1}$ . That is

$$(a_0) \subset \dots \subset (a_{n+1})$$

As  $R$  is Noetherian, this sequence terminates. Then  $a = a_nb_nb_{n-1} \dots b_1b_0$ . □

**Corollary 1.82.** If  $R$  is Noetherian, commutative then any non-unit  $a$ ,  $a \neq 0$  can be written as product of irreducibles:

$$a = a_1 \dots a_n$$

**Proof.** If  $a$  is irreducible we are done. Otherwise, we could write  $a = a_1b_1$ , for  $a_1$  irreducible. If  $b_1$  is irreducible we are done as well. Otherwise, write  $b_1 = a_2b_2$  and we could repeat this to give

$$a = a_1a_2 \dots a_{n-1}b_{n-1}$$

with  $a_i$  irreducible. We have a chain of ideal

$$(b_1) \subset (b_2) \subset \dots \subset (b_{n-1}) \subset (b_n)$$

As  $R$  is Noetherian, this chain terminates. Eventually,  $b_n$  is irreducible and

$$a = a_1 \dots a_{n-1}a_n, \quad a_n = b_n$$

with all  $a_i$  irreducible. □

**Remark.** In general, the decomposition is not unique. If it is unique, then  $R$  is called factorial domain. In particular, P.I.D. is factorial domain.

**Proposition 1.83.** If  $R$  is a commutative integral domain and  $p \in R$  prime. Then  $p$  is irreducible.

**Proof.** Suppose  $\square : R \rightarrow R/p$  be the conical mapping. So that  $R/p$  is an integral domain. Suppose  $p = bc$ , then

$$[b][c] = 0$$

Suppose  $[b] \neq 0$  then  $[c] = 0$  i.e.  $c = \lambda p$  for some  $\lambda$ . So  $p = bc = b\lambda p$ .  $R$  is an integral domain so  $1 = b\lambda$ . That is,  $b \in R^*$ . If  $[b] \neq 0$  then  $b \in R^*$ . Therefore,  $p$  is irreducible. □

**Lemma 1.84.** Let  $R$  be a P.I.D and let  $a \in R$  be irreducible. Suppose  $b \in R$  such that  $b \notin (a)$ . Then  $(a, b) = \{a\lambda + b\mu \mid \lambda, \mu \in R\} = R$ .

**Proof.** Consider the ideal  $(a, b) = \{a\lambda + b\mu \mid \lambda, \mu \in R\}$ . As  $R$  is P.I.D., there exists  $c \in R$  such that

$$(a, b) = (c)$$

Observe that  $a \in (c)$  so  $a = \lambda c$ . But  $a$  is irreducible so either  $\lambda \in R^*$  or  $c \in R^*$ . If  $\lambda \in R^*$  then  $c = \lambda^{-1}a$  so that  $(c) \subset (a) \subset (c)$ . But this implies  $(b) \subset (a) = (c)$ , a contradiction. Therefore, we must have  $c \in R^*$  so that  $(c) = R$ . In other words,  $(a, b) = R$ . □

**Corollary 1.85.** If  $R$  is P.I.D., every irreducible element is prime.

**Proof.** Take  $b, c \in R$ . Suppose  $[b][c] = 0$  where  $\square$  is the canonical mapping. Enough to show that if  $[b] \neq 0$  then  $[c] = 0$ . If  $[b] \neq 0$  then  $b \notin (a)$ , it follows that  $(a, b) = R$  i.e. there exists  $\lambda, \mu \in R$  such that

$$a\lambda + b\mu = 1$$

i.e.  $ac\mu + bc\mu = c$ . But since  $[b][c] = 0$  implies  $[bc] = 0$  so  $bc = a\nu$ . That is  $ac\lambda + a\mu\nu = c$ , so  $c \in (a)$  i.e.  $[c] = 0$ . □

**Remark.** In P.I.D., irreducible is equivalent to prime.

**Corollary 1.86.** If  $R$  is a commutative ring and  $a \in R \setminus \{0\}$  then

$$a = p_1 \cdots p_n$$

If  $p_i, p_j$  are distinct primes then

$$(p_i, p_j) = 1 = R$$

So if  $R$  is P.I.D.,  $a \in R \setminus \{0\}$ . We can write

$$a = up_1^{e_1} \cdots p_n^{e_n} \quad u \in R^*$$

where  $p_1, \dots, p_n$  are primes such that  $(p_i, p_j) = 1$  for  $i \neq j$ .

Let  $M$  be a finitely generated torsion module over P.I.D.  $R$ . Then there exists  $a \in R \setminus \{0\}$  such that  $M = \Theta_a(M) = \Theta_a$ . Now decompose  $a$  into product of prime powers i.e.  $a = p_1^{e_1} \cdots p_n^{e_n}$ . Then we know that

$$\Theta_a(M) = \Theta_{p_1^{e_1}}(M) \dot{+} \cdots \dot{+} \Theta_{p_n^{e_n}}(M).$$

First observation: If  $R$  P.I.D. and  $p \in R$  prime, then

$$\{\text{modules of exponent } e \text{ over } R\} \equiv \{\text{modules over } R/p^e\}.$$

Second observation: Let  $1 \leq d < e$ . Considered as a module over  $R/p^e$  we have  $H^1(R/p^1, R/p^e) = 0$ . Take  $e \geq 2$ ,  $1 \leq d < e$ , we have a free resolution of  $R/p^d$  over  $R$  as follows

$$0 \longrightarrow R \xrightarrow{p^d} R \xrightarrow{\varepsilon} R/p^d \longrightarrow 0$$

Comparing with

$$0 \longrightarrow R \xrightarrow{p^e} R/p^e \xrightarrow{\bar{\varepsilon}} R/p^d \longrightarrow 0$$

$$0 \longrightarrow R/p^d \longrightarrow R/p^e \xrightarrow{p^{e-d}} R/p^e \xrightarrow{\bar{\eta}} R/p^{e-d} \longrightarrow 0$$

We get an exact sequence

$$\cdots \longrightarrow R/p^e \xrightarrow{p^d} R/p^e \xrightarrow{p^{e-d}} R/p^e \xrightarrow{p^d} R/p^e \longrightarrow R/p^d \longrightarrow 0$$

Thus, over  $R/p^e$ , if  $1 \leq d < e$  then  $R/p^e$  has a periodic free resolution of infinite length.

**Corollary 1.87.** If  $1 \leq d < e$ , then  $H^k(R/p^d, R/p^e) = 0$  for all  $k$ .

**Proof.** Taking above free resolution and we want to compute  $H^1(R/p^d, R/p^e)$ . Notice

$$\text{Hom}_{R/p^e}(R/p^e, R/p^e) \cong R/p^e$$

So applying  $\text{Hom}(\cdot, R/p^e)$  to

$$0 \longrightarrow R/p^e \xrightarrow{p^d} R/p^e \xrightarrow{p^{e-d}} R/p^e \xrightarrow{p^d} R/p^e \longrightarrow \cdots$$

which is exact. So  $H^k(R/p^d, R/p^e) = 0$  for all  $k$ . In particular true for  $k = 1$ . □

**Corollary 1.88.** If  $1 \leq d \leq e$  then

$$H^k(R/p^d, R/p^e) = 0 \text{ for all } k \geq 1$$

**Proof.** Proved for  $1 \leq d < e$ , if  $d = e$  then  $R/p^e$  is free of rank 1 over  $R/p^e$ . □

Let  $R$  be P.I.D. and  $p$  prime. Define statement  $Q(e, m)$ : Let  $M$  be a torsion module with  $m$  generators and exponent at most  $e$  i.e.  $Mp^e = 0$ . Then  $M \cong \bigoplus R/p^{d_i}$  where each  $d_i \leq e$ .

**Proof.** Double induction.  $Q(e, 1)$  is true for all  $e$  and  $Q(1, m)$  is true for all  $m$ . If  $M$  is a module of exponent 1, then  $M$  is a vector space over  $R/p$  which is a field.  $M$  is a vector space over  $R/p$  field

$$M \cong R/p \oplus \cdots \oplus R/p.$$

Define

$$Q(e) = Q(e, 1) \wedge Q(e, 2) \wedge \cdots = \bigwedge_{i \geq 1} Q(e, i)$$

So  $Q(1)$  is true, we want to show  $Q(e, m) \implies Q(e, m+1)$ . Proof of  $Q(e, m) \implies Q(e, m+1)$ ,  $M$  has  $m+1$  generators  $e_1, \dots, e_{m+1}$ . And suppose each  $e_i p^e = 0$ . If each  $e_i$  satisfies  $e_i p^{e-1} = 0$  then by  $Q(e-1, m)$  the conclusion holds and  $M \cong \bigoplus R/p^{d_i}$  where  $d_i \leq e-1$ . If on the other hand there exists  $i$  such that  $\varphi_i p^{e-1} = 0$  where  $\varphi_{m+1} p^{e-1} \neq 0$  but  $\varphi_{m+1} p^e = 0$  so  $(\varphi_{m+1}) \cong R/p^e$ . We now have an exact sequence

$$0 \longrightarrow (\varphi_{m+1}) \longrightarrow M \longrightarrow M/(\varphi_{m+1}) \longrightarrow 0$$

$M/(\varphi_{m+1})$  has  $m$  generators. By induction hypothesis,

$$M/(\varphi_{m+1}) \cong \bigoplus R/p^{d_i}$$

where  $d_i \leq e$ . Then we know that  $(\varphi_{m+1}) \cong R/p^e$  so we have an exact sequence

$$0 \longrightarrow (\varphi_{m+1}) \longrightarrow M \longrightarrow M/(\varphi_{m+1}) \longrightarrow 0$$

i.e.

$$0 \longrightarrow R/p^e \longrightarrow M \longrightarrow \bigoplus R/p^{d_i} \longrightarrow 0$$

By earlier computation,

$$H^k(R/p^{d_i}, R/p^e) = 0$$

So  $H^k(\bigoplus R/p^{d_i}, R/p^e) = 0$  true for  $k = 1$ . By Baer's Theorem, sequence above splits so  $M \cong \bigoplus R/p^{d_i} \oplus R/p^e$  so that

$$Q(e-1) \wedge Q(e, m) \implies Q(e, m+1).$$

So that  $Q(e-1)$  implies  $Q(e)$  for all  $e$ . That is  $Q(e, 1)$  is true.  $\square$

**Proposition 1.89.** If  $R$  is a P.I.D. and  $p \in R$  is a prime. Then  $R/p$  is a field.

**Proof.** Suppose  $b \notin (p)$  we need to show that  $(b)$  is invertible in  $R/p$ . Earlier we proved that  $(p, b) = R$ . Choose  $\lambda, \mu$  such that

$$p\lambda + b\mu = 1$$

Reduce mod  $p$  we have  $0 \neq [b][\lambda] = 1$  so  $[b]$  is invertible i.e.  $R/p$  is a field.  $\square$

If  $M$  is finitely generated the

$$M \cong R^m \oplus \bigoplus R/p_i^{e_i}$$

where each  $p_i$  are primes and this is unique up to permutation.

**Example.** Let  $R = \mathbb{Z}$ , we get class function of finitely generated abelian groups  $A$  where

$$A \cong \mathbb{Z}^n \oplus \bigoplus \mathbb{Z}/p_i^{e_i}$$

**Example.** Let  $R = \mathbb{F}[t]$  where  $\mathbb{F}$  a field. If  $M$  is a finitely generated module over  $\mathbb{F}[t]$  then

$$M \cong \mathbb{F}[t]^m \oplus \mathbb{F}[t]/(q_i(t))^{e_i}$$

where each  $q_i(t)$  is irreducible polynomial. If  $\mathbb{F}$  is algebraically closed then  $q(t) = u(t - \lambda)$  which implies Jordan Normal Form Theorem.

## 1.13 06/03/2026

Let  $V$  be a vector space of  $\mathbb{F}$  and  $T : V \rightarrow V$  linear. Cayley-Hamilton:  $c_T(T) = 0$  where  $c_T(\lambda) = \det(T - \lambda I)$ .  $c_T(x) \in \mathbb{F}[x]$  we can regard  $(V, T)$  as torsion module over  $\mathbb{F}$ . Suppose

$$(V, T) \cong \bigoplus_{i=1}^N \mathbb{F}[x]/(q_i(x))^{e_i}$$

where  $q_i(x) \in \mathbb{F}[x]$  prime i.e. irreducible because  $\mathbb{F}[x]$  is P.I.D. and  $e_i \geq 1$ . In general that's best we can do. However, if  $\mathbb{F}$  is algebraically closed then  $q_i(x) = c_i(x - \lambda_i)$  for  $c_i \in \mathbb{F} \setminus \{0\}$  and  $\lambda_i \in \mathbb{F}$ . So if  $\mathbb{F}$  is algebraically closed, we could reduce to consider  $\mathbb{F}[x]/(x - \lambda)^n$ . As a vector space over  $\mathbb{F}$ ,  $\dim_{\mathbb{F}} \mathbb{F}[x]/(x - \lambda)^n = n$  with bases  $1, x, \dots, x^{n-1}$ . By an elementary change of basis, then  $1, (x - \lambda), \dots, (x - \lambda)^{n-1}$  is also a basis. If  $(V, T) \cong \mathbb{F}[x]/(x - \lambda)^n$ , there exists basis  $e_1 \rightarrow 1, e_2 \rightarrow (T - \lambda)e_1, e_3 \rightarrow (T - \lambda)e_2, \dots$ . So we get a basis  $e_1, \dots, e_n$  for  $V$  where  $e_r = (T - \lambda)^{r-1}e_1$ . Now calculate matrix of  $T$  with respect to this basis. Notice

$$T(e_1) = \lambda(e_1) + (T - \lambda)e_1 = \lambda e_1 + e_2$$

$$T(e_2) = \lambda e_2 + (T - \lambda)e_2 = \lambda e_2 + e_3$$

$$T(e_n) = \lambda e_n + (T - \lambda)e_n$$

where  $(T - \lambda)e_n = (T - \lambda)^n e_1 = 0$ . So we get our Jordan Normal Form.

**Theorem 1.90.** If  $T : V \rightarrow V$  is a linear map,  $V$  is finite dimensional vector space over  $\mathbb{F}$  where  $\mathbb{F}$  is algebraically closed. Then there exists a basis for  $V$  with respect to which matrix of  $T$  look like the Jordan block.

Let  $R$  be a commutative ring, suppose we have an exact sequence of chain-complexes over  $R$ :

$$0 \longrightarrow A_* \longrightarrow B_* \longrightarrow C_* \longrightarrow 0$$

$$\begin{array}{ccccccc}
0 & \longrightarrow & A_{n+1} & \xrightarrow{i} & B_{n+1} & \xrightarrow{p} & C_{n+1} \longrightarrow 0 \\
& & \downarrow \partial_{n+1}^A & & \downarrow \partial_{n+1}^B & & \downarrow \partial_{n+1}^B \\
0 & \longrightarrow & A_n & \xrightarrow{i} & B_n & \xrightarrow{p} & C_n \longrightarrow 0 \\
& & \downarrow \partial_n^A & & \downarrow \partial_n^B & & \downarrow \partial_n^B \\
0 & \longrightarrow & A_{n-1} & \xrightarrow{i} & B_{n-1} & \xrightarrow{p} & C_{n-1} \longrightarrow 0 \\
& & \downarrow & & \downarrow & & \downarrow \\
& & 0 & & 0 & & 0
\end{array}$$

By the algebraic MV theorem we have a long exact sequence

$$H_{n+1}(B) \xrightarrow{p} H_{n+1}(C) \xrightarrow{\partial} H_n(A) \xrightarrow{i} H_n(B) \xrightarrow{p} H_n(C) \xrightarrow{\delta} H_{n-1}(A)$$

The mapping cone construction is a special case: store with chain complexes  $(C_*, \gamma)$ ,  $(D_*, \delta)$ . Chain mapping  $f = (f_r) : (C_*, \partial^C) \rightarrow (D_*, \partial^D)$ . We have the following commutative diagram

$$\begin{array}{ccc}
C_{n+1} & \xrightarrow{f_{n+1}} & D_{n+1} \\
\downarrow \partial_{n+1}^C & & \downarrow \partial_{n+1}^D \\
C_n & \xrightarrow{f_n} & D_n \\
\downarrow \partial_n^C & & \downarrow \partial_n^D \\
C_{n-1} & \xrightarrow{f_{n-1}} & D_{n-1}
\end{array}$$

$f_* : C_* \rightarrow D_*$  given an exact sequence. Define  $M(f) = (M(f)_n, \partial_n^M)$  as follows

$$M(f)_n = D_n \oplus C_{n-1}$$

$$\partial_n^M : M(f)_n \rightarrow M(f)_{n-1}$$

$$D_n \oplus C_{n-1} \rightarrow D_{n-1} \oplus C_{n-2}$$

Let

$$\partial_n^M = \begin{pmatrix} \partial_n^D & (-1)^{n-1} f_{n-1} \\ 0 & \partial_{n-1}^C \end{pmatrix}$$

We claim that  $M(f) = (M(f)_n, \partial_n^M)$  is a chain-complex.

$$\partial_n^M \partial_{n+1}^M = \begin{pmatrix} \partial_n^D & (-1)^{n-1} f_{n-1} \\ 0 & \partial_{n-1}^C \end{pmatrix} \begin{pmatrix} \partial_{n+1}^D & (-1)^n f_n \\ 0 & \partial_n^C \end{pmatrix} = \begin{pmatrix} \partial_n^D \partial_{n+1}^D & (-1)^n \partial_n f_n + (-1)^{n-1} f_{n-1} \partial_n^C \\ 0 & \partial_{n-1}^C \partial_n^C \end{pmatrix} = \begin{pmatrix} 0 & (-1)^n \partial_n f_n + (-1)^{n-1} f_{n-1} \partial_n^C \\ 0 & 0 \end{pmatrix}$$

Commutativity implies  $(-1)^n \partial_n f_n + (-1)^{n-1} f_{n-1} \partial_n^C = 0$ . Thus,  $M(f)$  is definitely a chain complex. Let  $i_n : D_n \rightarrow M(f)_n$  by  $i_n(x) = \begin{pmatrix} x \\ 0 \end{pmatrix}$  and  $p_n : M(f)_n \rightarrow C_{n-1}$  where  $p_n \begin{pmatrix} x \\ y \end{pmatrix} = y$ . Define  $C_*(-1)$  to be the chain complex of  $C_*$  shifted given by 1 and

$$C_*(-1)_n = C_{n-1} \partial_n^{C_*(-1)} = \partial_{n-1}^C$$

We now have an exact sequence of chain complexes

$$0 \longrightarrow D_* \xrightarrow{i} M(f)_* \xrightarrow{p} C_*(-1) \longrightarrow 0$$

At this gives an long M-V sequence

$$H_{n+1}(D) \xrightarrow{i} H_{n+1}(M(f)) \xrightarrow{p} H_{n+1}(C_*(-1)) \xrightarrow{\delta} H_n(D) \longrightarrow H_n(M(f)) \longrightarrow H_n(C_*(-1))$$

$$H_{n+1}(D) \xrightarrow{i} H_{n+1}(M(f)) \xrightarrow{p} H_n(C_*) \xrightarrow{?} H_n(D_*) \xrightarrow{i} H_n(M(f)) \xrightarrow{p} H_{n-1}(C_*)$$

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Let  $f_* : C_* \rightarrow D_*$  be a chain mapping. And define  $M(f)_* = (M(f)_n, \partial_n^M)$ ,  $M(f)_n = D_n \oplus C_{n-1}$  and

$$\partial_n^M = \begin{pmatrix} \partial_n^D & (-1)^{n-1} f_{n-1} \\ 0 & \partial_{n-1}^C \end{pmatrix}$$

If

$$0 \longrightarrow A \xrightarrow{i} B \xrightarrow{p} C \longrightarrow 0$$

is an exact sequence of  $R$ -module. Let  $P_* \rightarrow A$ ,  $Q_* \rightarrow B$  be projective resolution. Then

$$\begin{array}{ccccccc}
0 & \longrightarrow & P_* & \xrightarrow{\hat{i}} & Q_* & & \\
& & \downarrow & & \downarrow & & \\
0 & \longrightarrow & A & \xrightarrow{i} & B & \xrightarrow{p} & C \longrightarrow 0
\end{array}$$

Suppose  $\hat{i} : P_* \rightarrow Q_*$  is a lifting of  $i$ .

**Theorem 1.91.** With the above notation. The mapping cone  $M(\hat{i})_*$  is a projective resolution of  $C$ .

**Proof.**  $M(f)_n = Q_n \oplus P_{n-1}$  is clearly projective. To show  $M(\hat{i})_*$  is a resolution. We need to show  $H_k(M\hat{i}) = 0$  for  $k \geq 1$ . We have an exact sequence of chain complexes

$$0 \longrightarrow Q_* \longrightarrow M(\hat{i})_* \longrightarrow P_{*-1} \longrightarrow 0$$

From the MV sequence, we have for each  $n \geq 2$  an exact sequence

$$H_n(Q_*) \longrightarrow H_n(M(\hat{i})_*) \xrightarrow{\delta} H_n(P_{*-1})$$

As  $P_*, Q_*$  are resolutions, they are exact so for  $n \geq 2$ .  $H_n(Q_*) = 0$ ,  $H_n(P_{*-1}) = H_{n-1}(P_*) = 0$ . So for  $n \geq 2$ ,  $H_n(M(\hat{i})_*) = 0$ . In low dimensions we get

$$\begin{aligned} H_1(Q_*) &\longrightarrow H_1(M(\hat{i})_*) \longrightarrow H_1(P_{*-1}) \xrightarrow{\delta} H_0(Q_*) \longrightarrow H_0(M(\hat{i})_*) \longrightarrow H_0(P_{*-1}) \\ 0 &\longrightarrow H_1(M(\hat{i})_*) \longrightarrow H_0(P_*) \xrightarrow{\pm i} H_0(Q_*) \longrightarrow H_0(M(\hat{i})_*) \longrightarrow 0 \end{aligned}$$

We have to interpret  $\delta : H_0(P_*) \rightarrow H_0(Q_*)$ . In terms of initial data  $H_0(P_*) \cong A$ ,  $H_0(Q_*) \cong B$

$$\begin{array}{ccccccc} 0 & \longrightarrow & Q_n & \longrightarrow & Q_n \oplus P_{n-1} & \longrightarrow & P_{n-1} \longrightarrow 0 \\ & & \downarrow \partial_n^Q & & \downarrow \partial_n^{M(\hat{i})} & & \downarrow \partial_{n-1}^{P_*} \\ 0 & \longrightarrow & Q_{n-1} & \longrightarrow & Q_{n-1} \oplus P_{n-2} & \longrightarrow & P_{n-2} \longrightarrow 0 \end{array}$$

where

$$\partial_n^{M(\hat{i})} = \begin{pmatrix} \partial_n^Q & (-1)^n \hat{i} \\ 0 & \partial_{n-1}^{P_*} \end{pmatrix}$$

We have  $H_1(M(\hat{i})_*) = 0$  so  $M((\hat{i}))_*$  is a resolution.

$$\begin{array}{ccccccc} 0 & \longrightarrow & H_0(P_*) & \xrightarrow{\pm i} & H_0(Q_*) & \longrightarrow & H_0(M(\hat{i})_*) \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & A & \xrightarrow{\pm i} & B & \longrightarrow & C \longrightarrow 0 \end{array}$$

□

We are trying to prove Hilbert Syzygy theorem:

**Theorem 1.92 (Hilbert Syzygy Theorem).** If  $A$  is a commutative ring, and  $A[t]$  be a polynomial ring of one variable. Then  $gd(A(t)) = gd(A) + 1$ . In particular

$$gd(A[t_1, \dots, t_n]) = gd(A) + n.$$

There are two steps: we need to show  $gd(R[t]) \leq gd(R) + 1$  and  $gd(R) + 1 \leq gd(R(t))$ . We shall demonstrate the proof of  $gd(R[t]) \leq gd(R) + 1$  first.

**Proof.** Let  $A$  be an  $R$ -module and let  $f : A \rightarrow A$  be an  $R$ -module homomorphism. Define  $R[t]$ -module  $A_f$ :

1. The underlying  $R$ -module structure of  $A_f$  is simply  $A$ .
2. The action  $R[t] \times A_f \rightarrow A_f$  is given by  $t \cdot a = f(a)$ .

We want to construct a  $R[t]$  projective resolution for  $A_f$ . Let  $P_* \rightarrow A$  be a projective resolution of  $A$  over  $R$ . Define  $A[t] = \bigoplus_{r=0}^{\infty} A \cdot t^r$  with finite support. There is an obvious projective resolution for  $A[t]$  over  $R[t]$  given by  $P_*(t)$  i.e.

$$P_n[t] = \bigoplus_{r=0}^{\infty} P_n t^r$$

$P_n[t] \rightarrow P_{n-1}[t]$  and  $\sum p_r t^r \mapsto \sum \partial(p_r) t^r$ .  $P_*(t)$  is a projective resolution of  $A(t)$  over  $R[t]$ . □

**Proposition 1.93.** There exists an exact sequence of  $R(t)$ -module

$$0 \longrightarrow A[t] \xrightarrow{j} A[t] \xrightarrow{\text{ev}} A_f \longrightarrow 0$$

where  $\text{ev}(\sum a_r t^r) = \sum f^r(a_r)$  and  $j = 1 \otimes t - f \otimes 1$  where  $(1 \otimes t)t^r = t^{r+1}$  and  $(f \otimes 1)(a_r t^r) = f(a_r)t^r$ .

**Proof.**  $\text{ev} : A[t] \rightarrow A_f$  is clearly surjective. And

$$\text{ev}(1 \otimes t)(\sum a_r t^r) = \text{ev}(\sum a_r t^{r+1}) = \sum f^{r+1}(a_r)$$

And

$$\text{ev}(f \otimes 1)(\sum a_r t^r) = \text{ev}(\sum f(a_r) t^r) = \sum f^{r+1}(a_r)$$

i.e.  $\text{ev}(1 \otimes t - f \otimes 1) = 0$ . So  $\text{Im}(1 \otimes t - f \otimes 1) \subset \ker(\text{ev})$ . To show exactness, we need to show  $\ker(\text{ev}) \subset \text{Im}(1 \otimes t - f \otimes 1)$ . We need the following Lemma. □

**Lemma 1.94.** For each  $n$ ,

$$(1 \otimes t^n - f^n \otimes 1)(a) \in \text{Im}(1 \otimes t - f \otimes 1)$$

**Proof.** By induction on  $n$ . For  $n = 1$ :

$$(1 \otimes t - f \otimes 1)(a) \in \text{Im}(1 \otimes t - f \otimes 1)$$

Suppose true for  $n - 1$ :

$$\begin{aligned} (1 \otimes t^n - f^n \otimes 1)(a) &= (1 \otimes t)(1 \otimes t^{n-1})(a) - (f \otimes 1)(f^{n-1} \otimes 1)(a) + (1 \otimes t)(f^{n-1} \otimes 1)(a) - (1 \otimes t)(f^{n-1} \otimes 1)(a) \\ &= (1 \otimes t)(1 \otimes t^{n-1} - f^{n-1} \otimes 1)(a) + (1 \otimes t - f \otimes 1)(f^{n-1} \otimes 1)(a) \\ &= (1 \otimes t^{n-1} - f^{n-1} \otimes 1)(1 \otimes t)(a) + (1 \otimes t - f \otimes 1)(f^{n-1} \otimes 1)(a) \end{aligned}$$

By induction, the first term is in the image and obviously the second term is in the image.  $\square$

So we have the following data

$$\begin{array}{ccccccc} 0 & \longrightarrow & P_*[t] & \xrightarrow{\hat{j}} & P_*[t] & & \\ & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & A_*[t] & \xrightarrow{j} & A_*[t] & \xrightarrow{\text{ev}} & A_f \longrightarrow 0 \end{array}$$

The bottom row is exact. The conclusion is that  $M(\hat{j})$  is a projective resolution of  $A_f$ . Let  $A$  be an  $R$ -module,  $f : A \rightarrow A$  is a  $R$ -module. We define  $A_f$  as an  $R[t]$ -module, we get an exact sequence

$$0 \longrightarrow A[t] \xrightarrow{j} A[t] \xrightarrow{\text{ev}} A_f \longrightarrow 0$$

is a projective resolution  $P_*(t) \rightarrow A[t]$ .

**Proposition 1.95.**  $cd_{R[t]}(A_f) \leq cd_R(A) + 1$ .

**Proof.** Suppose  $cd_R(A) = n$ , let  $P_* \rightarrow A$  be a projective resolution of length  $n$ .  $M(\hat{f})$  is a projective resolution of  $A_f$  so  $M(\hat{j})_{n+2} = P[t]_{n+2} \oplus P[t]_{n+1} = 0$ .  $M(\hat{j})_{n+1} = P[t]_{n+1} \oplus P[t]_n = P[t]_n$ . Therefore,  $M(\hat{j})_{n+1} \neq 0$  so  $cd_{R[t]}(A_f) \leq n + 1 = cd_R(A) + 1$ .  $\square$

As  $cd_R(A) \leq gd(R)$  then  $cd_{R[t]}(A_f) \leq gd(R) + 1$ . Any  $R[t]$ -module can be represented in the form  $A_f$  for some  $R$ -module  $A$  and some  $R$ -linear  $f : A \rightarrow A$ . Let  $\mathcal{A}$  be an  $R[t]$ -module, let  $A$  be underlying  $R$ -module and let  $f : A \rightarrow A$  be multiplication by  $t$  then  $\mathcal{A} \cong A_f$

**Corollary 1.96.** If  $\mathcal{A}$  is any  $R[t]$ -module. Then  $cd_{R[t]}(\mathcal{A}) \leq gd(R) + 1$ .

So we have

**Theorem 1.97.**  $gd(R[t]) \leq gd(R) + 1$ .

It remains to show  $gd(R) + 1 \leq gd(R[t])$ . Suppose  $gd(R) = n$  so there exists  $R$ -module  $A, B$  such that  $H^n(A, B) \neq 0$ . We will produce  $R[t]$ -module  $\hat{A}, \hat{B}$  such that  $H^{n+1}(\hat{A}, \hat{B}) \cong H^n(A, B)$  so  $cd(A) + 1 \leq cd(\hat{A})$ . Definition of  $\hat{A}$  Let  $A$  be an  $R$ -module  $\hat{A} = A$  be an  $R$ -module and the action of  $R[t]$  on  $\hat{A}$  is that  $t$  acts on zero. As before, define

$$A[t] = \bigoplus_{r=0}^{\infty} A \cdot t^r.$$

There is now an obvious exact sequence

$$0 \longrightarrow A[t] \xrightarrow{\times t} A[t] \xrightarrow{\varepsilon} \hat{A} \longrightarrow 0$$

where  $\varepsilon(\sum a_r t^r) = a_0$ . Put  $n = cd_R(A)$ , let  $P_* \rightarrow A$  be a projective resolution of length  $n$  over  $R$  and  $P_*[t] \rightarrow A[t]$  is a projective resolution of length  $n$  over  $R[t]$ .

$$\begin{array}{ccccccc} 0 & \longrightarrow & P_*[t] & \xrightarrow{\times t} & P_*[t] & & \\ & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & A[t] & \xrightarrow{\times t} & A[t] & \xrightarrow{\varepsilon} & \hat{A} \longrightarrow 0 \end{array}$$

By Mapping Cone Theorem,  $M(t)$  is a projective resolution of  $\hat{A}$  and it has length  $n+1$ .  $M(t)_{n+2} = 0$ ,  $M(t)_{n+1} = 0 \oplus P_n[t]$ ,  $M(t)_n = P_n[t] \oplus P_{n-1}[t]$ . Use  $M(t)$  to compute  $H^{n+1}(\hat{A}, \hat{B})$  and recall that  $\hat{A}, \hat{B}$  and  $t$  acts on zero.

$$\begin{array}{ccccccc} 0 & \longrightarrow & M(t)_{n+1} & \longrightarrow & M(t)_n & \longrightarrow & M(t)_{n-1} \\ & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & 0 \oplus P_n[t] & \xrightarrow{\partial_{n+1}} & P_n[t] \oplus P_{n-1}[t] & \xrightarrow{\partial_n} & P_{n-1}[t] \oplus P_{n-2}[t] \end{array}$$

To compute  $H^{n+1}(\hat{A}, \hat{B})$  applying  $\text{Hom}_{R[t]}(\cdot, \hat{B})$ :

$$\text{Hom}(P_n[t], \hat{B}) \oplus \text{Hom}(P_{n-1}[t], \hat{B}) \longrightarrow \text{Hom}(P_n[t], \hat{B}) \longrightarrow 0$$

As  $t$  acts trivially on  $\hat{B}$ , we get

$$\text{Hom}_{R[t]}(P_n[t], \hat{B}) = \text{Hom}_R(P_n, B)$$

So we can replace the above sequence by

$$\text{Hom}_R(P_n, B) \oplus \text{Hom}_R(P_{n-1}, B) \longrightarrow \text{Hom}_R(P_n, B) \longrightarrow 0$$

$$H^{n+1}(\hat{A}, \hat{B}) \cong \text{Hom}_R(P_n, B) / \text{Im}(\partial : \text{Hom}(P_{n-1}, B) \rightarrow) \cong H^n(A, B)$$

A module over  $R$  with  $cd(R) = n$ ,  $H^n(A, B) \neq 0$  then  $\hat{A}$  is a module over  $R[t]$  so  $H^{n+1}(\hat{A}, \hat{B}) \neq 0$  which certainly implies that

$$cd_{R[t]}(\hat{A}) \geq n + 1 = cd_R(A) + 1$$

If  $A$  is chosen to be of maximal then  $cd_R(A) = gd(R)$  where we have produced an  $R[t]$ -module  $\hat{A}$  such that  $cd_{R[t]}(\hat{A}) \geq gd(R) + 1$ . So  $gd(R[t]) \geq gd(R) + 1$ .

**Remark.**  $gd(R[t_1, \dots, t_n]) = gd(R) + n$ .

**Example.** If  $R = \mathbb{Z}$ ,  $gd(\mathbb{Z}[t]) = 1 + 1 = 2$ . Since every P.I.D. has  $gd = 1$  so  $\mathbb{Z}[t]$  is not a P.I.D.

## 1.15 13/03/2026

Let  $f : C_* \rightarrow D_*$  and  $M(f)$  is projective with  $\text{coker}(H_0(C) \rightarrow H_0(D))$ . Compute a free resolution of  $\mathbb{Z}$  over  $\mathbb{Z}[x, y]$ . We know a resolution for  $\mathbb{Z}$  over  $\mathbb{Z}[x]$ .

$$0 \longrightarrow \mathbb{Z}[x] \xrightarrow{\times x} \mathbb{Z}[x] \longrightarrow \mathbb{Z} \longrightarrow 0$$

And we also have

$$\begin{array}{ccccccc} & & 0 & & & & \\ & & \downarrow & & & & \\ & C_1 = \mathbb{Z}[x, y] & \xrightarrow{\times y} & \mathbb{Z}[x, y] = D_1 & & & \\ & \downarrow \times x & & \downarrow \times x & & & \\ & C_0 = \mathbb{Z}[x, y] & \xrightarrow{\times y} & \mathbb{Z}[x, y] = D_0 & & & \\ & \downarrow \varepsilon \otimes 1 & & \downarrow \varepsilon \otimes 1 & & & \\ 0 & \longrightarrow & \mathbb{Z}[y] & \xrightarrow{\times y} & \mathbb{Z}[y] & \longrightarrow & \mathbb{Z} \longrightarrow 0 \end{array}$$

Now take  $M(\times y)$ ,  $M_2 = D_2 \oplus C_1 = 0 \oplus C_1$ ,  $M_1 = D_1 \oplus C_0$  and  $M_0 = D_0 \oplus C_{-1} = D_0 \oplus 0 = D_0$ . That is, we have

$$C_1 \xrightarrow{\partial_2^M} D_1 \oplus C_0 \xrightarrow{\partial_1^M} D_0$$

where

$$\partial_2^M = \begin{pmatrix} \partial_2^D & (-1)y \\ 0 & \partial_1^C \end{pmatrix} = \begin{pmatrix} -y \\ 0 \end{pmatrix}, \quad \partial_1^M = (\partial_1^D \quad y) = (x, y)$$

Then

$$0 \longrightarrow \mathbb{Z}[x, y] \xrightarrow{\partial_2} \mathbb{Z}[x, y] \oplus \mathbb{Z}[x, y] \xrightarrow{\partial_1} \mathbb{Z}[x, y] \longrightarrow \mathbb{Z} \longrightarrow 0$$

This is known as Koszul resolution. Dualise i.e. take transpose

$$0 \longrightarrow F_0^* \xrightarrow{\partial_1^*} F_1^* \longrightarrow F_2^* \xrightarrow{\partial_2^*} 0$$

which means

$$0 \longrightarrow \mathbb{Z}[x, y] \xrightarrow{\partial_1^*} \mathbb{Z}[x, y] \oplus \mathbb{Z}[x, y] \xrightarrow{\partial_2^*} \mathbb{Z}[x, y]$$

$K^*$  and  $K$  are isomorphic after "change of variables".

**Corollary 1.98.** If  $\Lambda = \mathbb{Z}[x, y]$  then

$$H_\Lambda^k(\mathbb{Z}, \Lambda) = \begin{cases} 0 & k \neq 2 \\ 2 & k = 2 \end{cases}$$

Using this, find a resolution of  $\mathbb{Z}$  over  $\mathbb{Z}[x, y, z]$

$$\begin{array}{ccccccc}
& & 0 & & 0 & & \\
& & \downarrow & & \downarrow & & \\
& & C_2 & \xrightarrow{\times z} & D_2 & & \\
& & \downarrow \partial_2^C & & \downarrow \partial_2^D & & \\
& & C_1 & \xrightarrow{\times z} & D_1 & & \\
& & \downarrow \partial_1^C & & \downarrow \partial_1^D & & \\
& & C_0 & \xrightarrow{\times z} & D_0 & & \\
& & \downarrow & & \downarrow & & \\
0 & \longrightarrow & \mathbb{Z}[z] & \xrightarrow{\times z} & \mathbb{Z}[z] & \longrightarrow & \mathbb{Z} \longrightarrow 0
\end{array}$$

Then

$$\begin{aligned}
M_3 &= D_3 \oplus C_2 = 0 \oplus C_2 = C_2 = \mathbb{Z}[x, y, z] \\
M_2 &= D_2 \oplus C_1 = \mathbb{Z}[x, y, z] \oplus \mathbb{Z}[x, y, z]^{(2)} = \mathbb{Z}[x, y, z]^{(3)} \\
M_1 &= D_1 \oplus C_0 = \mathbb{Z}[x, y, z]^{(2)} \oplus \mathbb{Z}[x, y, z] = \mathbb{Z}[x, y, z]^{(3)} \\
M_0 &= D_0 \oplus C_{-1} = \mathbb{Z}[x, y, z] \oplus 0 = \mathbb{Z}[x, y, z]
\end{aligned}$$

Then

$$\partial_3^M = \begin{pmatrix} \partial_3^D & z \\ 0 & \partial_2^C \end{pmatrix} = \begin{pmatrix} z \\ -y \\ z \end{pmatrix}, \quad \partial_2^M = \begin{pmatrix} \partial_2^D & -z \\ 0 & \partial_1^C \end{pmatrix} = \begin{pmatrix} -y & -z & 0 \\ x & 0 & -z \\ 0 & x & y \end{pmatrix}, \quad \partial_1^M = (\partial_1^D \quad +z) = (x \quad y \quad z)$$

## 1.16 16/03/2026

Let

$$0 \longrightarrow K \xrightarrow{i} P \xrightarrow{p} M \longrightarrow 0$$

be projective.

**Theorem 1.99** (Swan's Theorem). If  $f : M \rightarrow M$  is such that  $f^* : H^k(M, N) \rightarrow H^k(M, N)$  is an isomorphism for  $k = 1, 2$  and all  $N$ . Then  $\underline{\lim}(p, f)$  is projective.

**Proof.** Apply  $\text{Hom}(\cdot, N)$ , and take the cohomology sequence.

$$\begin{array}{ccccccccc}
\text{Hom}(K, N) & \longrightarrow & H^1(M, N) & \longrightarrow & H^1(P, N) & \longrightarrow & H^1(K, N) & \longrightarrow & H^2(M, N) \\
\downarrow \text{id}^* & & \downarrow f^* & & \downarrow \pi_1^* & & \downarrow \text{id}^* & & \downarrow f^* \\
\text{Hom}(K, N) & \longrightarrow & H^1(M, N) & \longrightarrow & H^1(\underline{\lim}, N) & \longrightarrow & H^1(K, N) & \longrightarrow & H^2(M, N)
\end{array}$$

By hypothesis,  $\text{id}^*$  and  $f^*$  are all isomorphisms. That is, by the Five Lemma,  $H^1(\underline{\lim}, N) \cong H^1(P, N)$ . As  $P$  is projective, then  $H^1(P, N) = 0$  i.e.  $H^1(\underline{\lim}, N) = 0$  for all  $N$  i.e.  $\underline{\lim}(p, f)$  is projective.  $\square$

Question: What does  $f$  look like if  $f^* : H^k(M, N) \rightarrow H^k(M, N)$  is isomorphism for all  $k$ .

Slightly different question: Given  $g : M \rightarrow M$ , when is  $g^* : H^k(M, N) \rightarrow H^k(M, N)$  the zero map. Suppose

$$\begin{array}{ccc}
M & \xrightarrow{g} & M \\
\downarrow \eta & \nearrow \xi & \\
Q & & 
\end{array}$$

$g = \xi \circ \eta$  since  $Q$  is projective. In this case,  $g^* : H^k(M, N) \rightarrow H^k(M, N)$  is zero for all  $k \geq 1$  and all  $N$ .

**Proof.**  $g^* = \eta^* \circ \xi^*$ ,  $\xi^* : H^k(M, N) \rightarrow H^k(M, N)$ . But  $H^k(Q, N) = 0$  since  $Q$  is projective. So  $\xi^* = 0$  so  $g^* = 0$ .  $\square$

**Proposition 1.100.** Let  $\alpha, \beta : M \rightarrow M$  and suppose that  $\alpha - \beta$  factors through a projective. Then  $\alpha^* = \beta^* : H^k(M, N) \rightarrow H^k(M, N)$  for all  $k \geq 1$  and for all  $N$ .

**Proof.**  $(\alpha - \beta)^* = 0$  so  $\alpha^* = \beta^*$ .  $\square$

Sufficient condition for  $f^* : H^k(M, N) \rightarrow H^k(M, N)$  to be an isomorphism for  $k \geq 1$  and all  $N$ . When there exists actual isomorphism  $h : M \rightarrow M$  such that  $h - f$  factors through projective, then  $f^* = h^* : H^k(M, N) \rightarrow H^k(M, N)$ .

**Example.** Let  $R$  be an integral domain of characteristic zero. Let  $G$  be a finite group.  $R[G]$  group ring of  $G$  over  $R$

where

$$R[G] = \left\{ \sum_{g \in G} a_g g : a_g \in R \right\}$$

Let  $\varepsilon(g) = 1$  for all  $g$  and  $\varepsilon(\sum_{g \in G} a_g g) = \sum_{g \in G} a_g$ .

**Proposition 1.101.** If  $h : R[G] \rightarrow R$  is an  $R[G]$ -homomorphism then  $h = c\varepsilon$  for  $c \in R$ .

**Proof.**  $G$  acts trivially on  $R$  i.e.

$$\begin{aligned} h : R[G] &\rightarrow R \\ h(g) &= h(1 \cdot g) = h(1) \cdot g \end{aligned}$$

i.e.  $h$  takes same values on each basis element. So

$$h(\sum a_g g) = \sum a_g h(g) = (\sum a_g) h(1) = h(1) \varepsilon(\sum a_g g)$$

i.e.  $h = h(1)\varepsilon$ . □

How about  $h : R \rightarrow R[G]$ ?  $h$  completely determined by  $h(1)$ .  $h$  completely determined by  $h(1)$  so suppose

$$h(1) = \sum_{g \in G} a_g g$$

Let  $\gamma \in G$  then

$$h(1)\gamma^{-1} = \sum_{g \in G} a_g (g\gamma^{-1}) = \sum_{\delta \in G} a_{\delta\gamma} \delta$$

Suppose  $\delta\gamma = 1$  then  $a_1$  is the coefficient of  $\delta$  and that is true for all  $\delta$ . So if  $h : R \rightarrow R[G]$ ,  $h(1) = \sum a_g g$ . Then  $a_g = a_1$  for all  $g$ .

$$h = a_1 \left( \sum_{g \in G} g \right)$$

Define  $\varepsilon^* : R \rightarrow R[G]$  then  $\varepsilon^* = \sum_{g \in G} g$ .

**Proposition 1.102.** If  $h : R \rightarrow R[G]$  is an  $R[G]$ -homomorphism then  $h = c\varepsilon^*$  where  $\varepsilon^*(1) = \sum_{g \in G} g$ .

**Proposition 1.103.**  $\varepsilon \circ \varepsilon^* = |G|$ .

**Proof.**  $\varepsilon \circ \varepsilon^*(1) = \varepsilon(\sum_{g \in G} g) = \sum_{g \in G} \varepsilon(g)$ . But each  $\varepsilon(g) = 1$  so

$$\varepsilon \circ \varepsilon^*(1) = |G|$$

□

**Proposition 1.104.** Let  $G$  be a finite group and  $R$  be an integral domain with  $\text{char} = 0$ . Let  $f : R \rightarrow R$  be an  $R[G]$ -homomorphism which factors through a projective then  $f(1) = \lambda|G|$ .

**Proof.** We are assuming that  $f$  factors then

$$\begin{array}{ccc} M & \xrightarrow{f} & M \\ \downarrow \eta & \nearrow \xi & \\ P & & \end{array}$$

with  $P$  projective.

Case 1:  $P = R[G]$ ,

$$\begin{array}{ccc} R & \xrightarrow{f} & R \\ \downarrow b\varepsilon^* \nearrow a\varepsilon & & \\ R[G] & & \end{array}$$

$$f(1) = (a\varepsilon)(b\varepsilon^*)(1) = (a\varepsilon)\varepsilon\varepsilon^*(1) = ab|G| \text{ i.e. } \lambda = ab.$$

Case 2:  $P = R[G]^{(n)} = \underbrace{R[G] \oplus \cdots \oplus R[G]}_n$ .

$$\begin{array}{ccc} R & \xrightarrow{f} & R \\ \downarrow b\varepsilon^* \nearrow a\varepsilon & & \\ R[G] \oplus \cdots \oplus R[G] & & \end{array}$$

then

$$\eta = \begin{pmatrix} a_1 \varepsilon^* \\ \vdots \\ a_n \varepsilon^* \end{pmatrix}$$

and  $\xi = (b_1 \varepsilon, \dots, b_n \varepsilon)$ .

$$(\xi \circ \eta)(1) = \sum a_i b_i \varepsilon \varepsilon^*(1) = \left( \sum a_i b_i \right) |G|$$

Case 3: If  $P$  is finitely generated projective,  $P \oplus Q = R|G|^n$ :

$$\begin{array}{ccc} R & \xrightarrow{f} & R \\ \downarrow b\varepsilon^* & \nearrow a\varepsilon & \\ P & & \end{array}$$

Define  $\hat{\eta} : R \rightarrow P \oplus G$  by  $\hat{\eta}(1) = (\eta(1), 0)$ . Then  $f(1) = \lambda|G|$  for some  $\lambda$ . □

Let  $G$  be a finite group,  $R$  be an integral domain of char  $\neq 0$ . Let

$$\begin{array}{ccccccc} 0 & \longrightarrow & I & \longrightarrow & \varinjlim(f) & \longrightarrow & R \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & I & \longrightarrow & R|G| & \xrightarrow{\varepsilon} & R \longrightarrow 0 \end{array}$$

By Swan's Theorem,  $\varinjlim(\varepsilon, f)$  is projective if  $f : R \rightarrow R$  induces  $f^* : H^k(R, N) \rightarrow H^k(R, N)$ .  $f$  is completely determined by  $f(1)$ .  $\text{ev} : \text{Hom}_{R|G|}(R, R) \rightarrow R$ ,  $\text{ev}(f) = f(1)$  is an isomorphism.  $f : R \rightarrow R$  is an actual isomorphism if  $f(1) \in R^*$  (unit group). Identify  $\text{Hom}_{R|G|}(R, R)$  with  $R$ . Let  $(|G|) \subset R$  be the ideal generated by  $|G|$ . In particular,  $R/|G|$  is the quotient ring.

**Proposition 1.105.** If  $f : R \rightarrow R$  maps to  $\text{id} \in R/|G|$  then  $f : H^k(R, N) \rightarrow H^k(R, N)$  is an isomorphism for all  $k \geq N$ .

**Proof.**  $f$  defines  $1_{R/|G|}$  if and only if  $f - \text{id} \in |G|$  i.e.  $f - \text{id}$  factors through projective. □

**Example.** Take  $G = C_4 = \{t \mid t^4 = 1\}$ , take  $R = \mathbb{Z}[x]$  polynomials in 1-variable over  $\mathbb{Z}$ .  $R/|G| = (\mathbb{Z}/4)[x]$ .

**Proposition 1.106.**  $(\mathbb{Z}/4)[x]^*$  has infinitely many elements of order 2.

**Proof.** Let  $u = 1 + 2p(x) \in R/4$  then  $u^2 = (1 + 2p(x))(1 + 2p(x)) = 1 + 4p(x) + 4p(x)^2 = 1$ . □

**Corollary 1.107.** Let  $R = \mathbb{Z}[x]$ ,  $G = C_4$  then  $R|C_4|$  has infinitely many projective.

**Proof.** Take

$$\begin{array}{ccccccc} 0 & \longrightarrow & I & \longrightarrow & \varinjlim & \longrightarrow & R \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow 1+2p(t) \\ 0 & \longrightarrow & I & \longrightarrow & R|C_4| & \longrightarrow & R \longrightarrow 0 \end{array}$$

□

**Example (Small Case).**  $G = Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$  where  $i^2 = j^2 = k^2 = -1$ ,  $ijk = 1$ . Take

$$\begin{array}{ccccccc} 0 & \longrightarrow & I & \longrightarrow & \varinjlim & \xrightarrow{\varepsilon} & \mathbb{Z} \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \times 3 \\ 0 & \longrightarrow & I & \longrightarrow & \mathbb{Z}|Q_8| & \xrightarrow{\varepsilon} & \mathbb{Z} \longrightarrow 0 \end{array}$$

3 is not an isomorphism over  $\mathbb{Z}|Q_8|$  but it is an isomorphism over  $\mathbb{Z}/8$ .

**Example.** Let  $G$  be a finite multiplicative abelian group. If  $|G| = p^n$  with  $m \geq 2$  then  $R|G|$  has infinitely many projectives which are not free.

Derived module category: Let  $\Lambda$  be a ring,  $\text{Mod}_\Lambda$  is finitely generated  $\Lambda$ -modules,  $\text{Der}(\Lambda)$  has the same objects as  $\text{Mod}_\Lambda$ . To define  $\text{Hom}_{\text{Der}}(M, N)$ , define  $\text{Hom}_0(M, N) = \{f : M \rightarrow N : f = \xi \circ \eta\}$  where  $f$  factors through projective. Then

$$\text{Hom}_{\text{Der}}(M, N) = \text{Hom}_\Lambda(M, N) / \text{Hom}_0(M, N)$$

i.e. factor out morphism which factors through projective. Paraphrase:

$$\text{Der}(\Lambda) = \text{Mod}_\Lambda / P \equiv 0$$

if  $P$  is projective. Let

$$\begin{array}{ccccccc} 0 & \longrightarrow & K & \xrightarrow{i} & P & \xrightarrow{p} & M \longrightarrow 0 \\ & & \downarrow f & & \downarrow \nu & & \downarrow \text{id} \\ 0 & \longrightarrow & K & \longrightarrow & \underline{\lim}(f, i) & \longrightarrow & M \longrightarrow 0 \end{array}$$

This works over  $\Lambda = R|G|$ ,  $\underline{\lim}(f, i)$  is projective implies  $f$  defines an isomorphism in  $\text{Der}$ .

**1.17 20/03/2026**